Physical Chemistry (I)

Lecture 20

Phase Diagrams

In the Last Class

- Solution models
  - Ideal-dilute solutions
  - Regular solutions
  - Bragg-Williams model
- Deviations from ideality
  - Excess function
  - Activity
  - Activity of BW model

Last class

Ideal-Dilute Solution

- Ideal-dilute solutions: mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law

- Solvent: Raoult's law ($p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$)

$$\mu_{\mathrm{A}}(l) = \mu_{\mathrm{A}}^{*}(l) + RT \ln x_{\mathrm{A}}$$

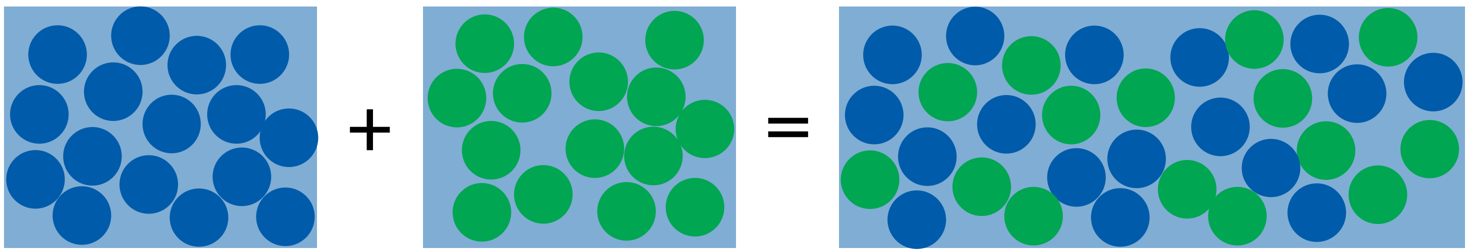
- Solute: Henry's law ($p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$)

$$\mu_{\mathrm{B}}(l) = \mu_{\mathrm{B}}^{\ominus}(l) + RT \ln x_{\mathrm{B}}$$

Last class

Regular Solution

Pure substances Mixture



A model solution with

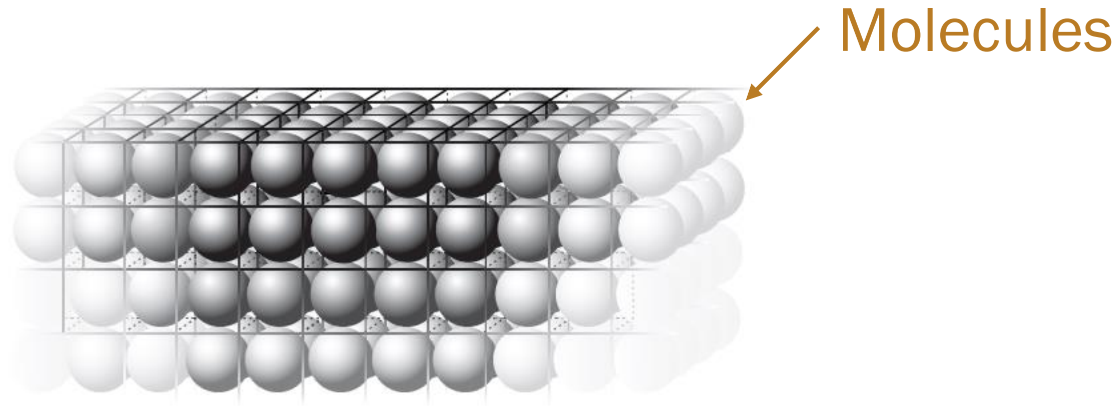
$$\Delta_{\text{mix}}S = \Delta_{\text{mix}}S^{\text{ideal}} = -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

and

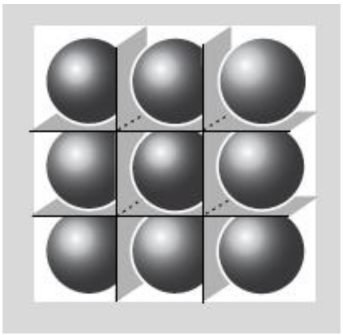
$$\Delta_{\text{mix}}H \neq 0$$

Last class

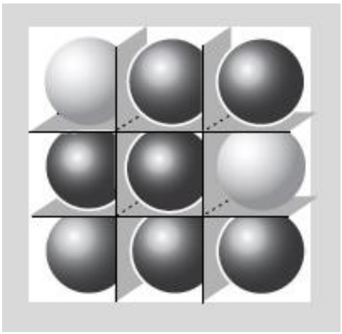
Bragg-Williams Model



Pure substance



Mixture



(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 14.1 and 16.2)

Last class

Bragg-Williams Model

$$\Delta_{\text{mix}} S = -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

$$\Delta_{\text{mix}} U = \xi nRT x_{\text{A}} x_{\text{B}}$$

$$\Delta_{\text{mix}} A = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}} + \xi x_{\text{A}} x_{\text{B}})$$

Last class

Deviations from Ideality

- Excess function

$$X^{\mathrm{E}} = \Delta_{\mathrm{mix}}X - \Delta_{\mathrm{mix}}X^{\mathrm{ideal}}$$

- Activity

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT\ln a_{\mathrm{A}}$$

Activity a_J

- Ideal-dilute solution
  - Solvent: $\mu_{\mathrm{A}}(l) = \mu_{\mathrm{A}}^{*}(l) + RT \ln x_{\mathrm{A}}$
  - Solute: $\mu_{\mathrm{B}}(l) = \mu_{\mathrm{B}}^{\ominus}(l) + RT \ln x_{\mathrm{B}}$

    where $\mu_{B}^{\ominus}(l) = \mu_{\mathrm{B}}^{*}(l) + RT \ln(K_{\mathrm{B}}/p_{\mathrm{B}}^{*})$

- Real solution
  - Solvent: $\mu_{\mathrm{A}}(l) = \mu_{\mathrm{A}}^{*}(l) + RT \ln a_{\mathrm{A}}$
  - Solute: $\mu_{\mathrm{B}}(l) = \mu_{\mathrm{B}}^{\ominus}(l) + RT \ln a_{\mathrm{B}}$

Last class

Activity Coefficient

$$\gamma_J = a_J / x_J$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A = \mu_A^*(l) + RT \ln x_A + RT \ln \gamma_A$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^\ominus(l) + RT \ln x_B + RT \ln \gamma_B$$

As $x_A \to 1$ and $x_B \to 0$, $a_J \to x_J$ and $\gamma_J \to 1$

Activity of BW Model

$$x_{\mathrm{A}} \ln \gamma_{\mathrm{A}} + x_{\mathrm{B}} \ln \gamma_{\mathrm{B}} = \xi x_{\mathrm{A}} x_{\mathrm{B}}$$

The following equations satisfy the given criterion:

$$\ln \gamma_{\mathrm{A}} = \xi x_{\mathrm{B}}^2, \text{ or } \gamma_{\mathrm{A}} = e^{\xi x_{\mathrm{B}}^2}$$

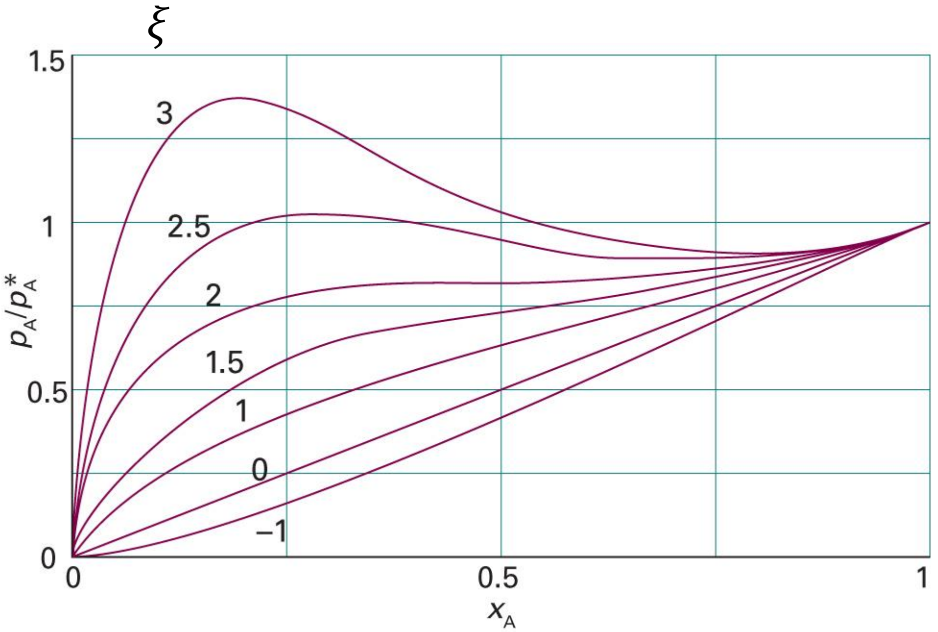
$$\ln \gamma_{\mathrm{B}} = \xi x_{\mathrm{A}}^2, \text{ or } \gamma_{\mathrm{B}} = e^{\xi x_{\mathrm{A}}^2}$$

(Margules equations)

Last class

Vapor Pressure of BW Model

$$p_{\mathrm{A}}/p_{\mathrm{A}}^{*} = x_{\mathrm{A}} e^{\xi(1-x_{\mathrm{A}})^2}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5F.2)

Phase Transitions of Mixtures?

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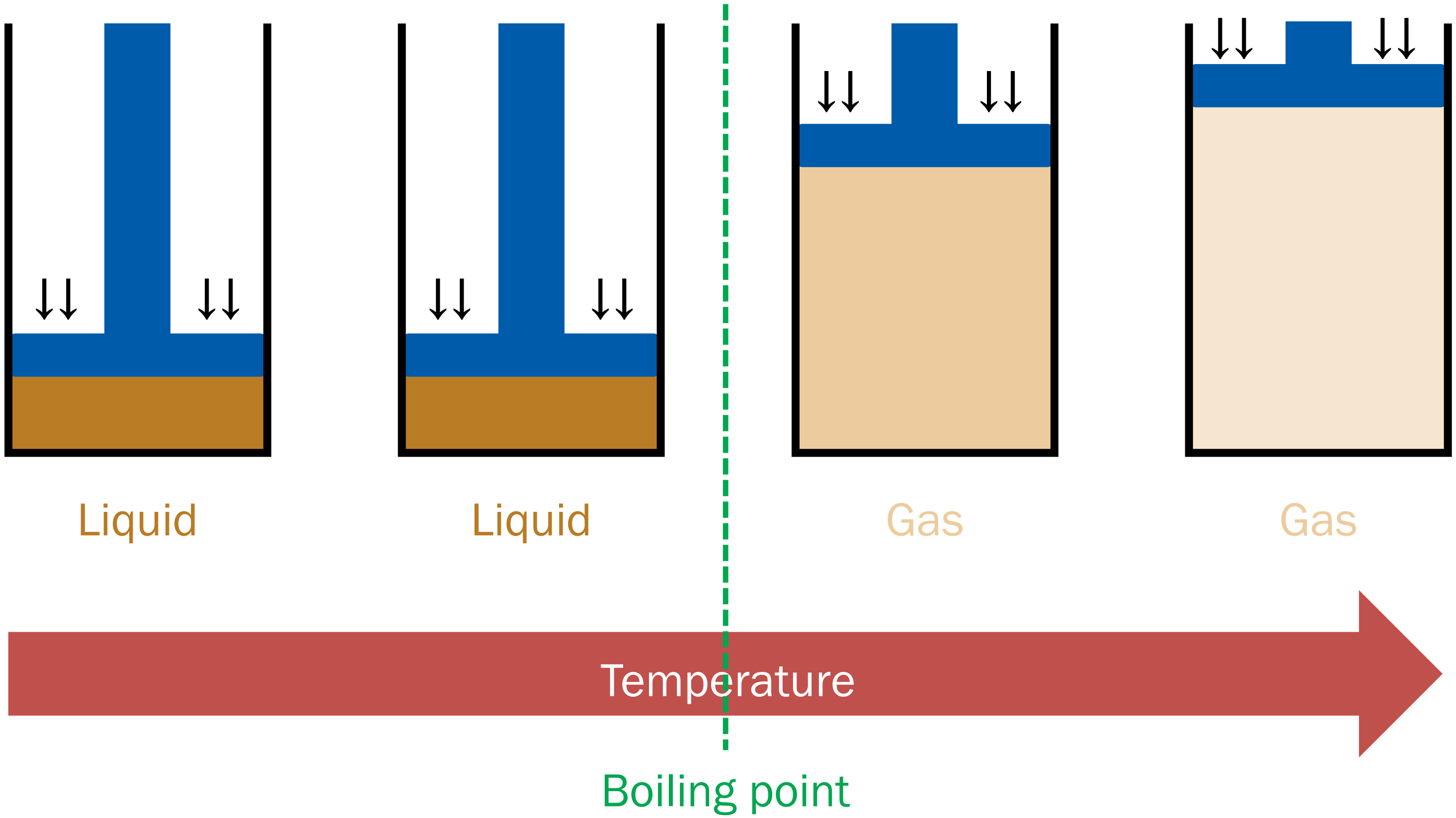
Boiling of Pure Substance

(Isobaric change)



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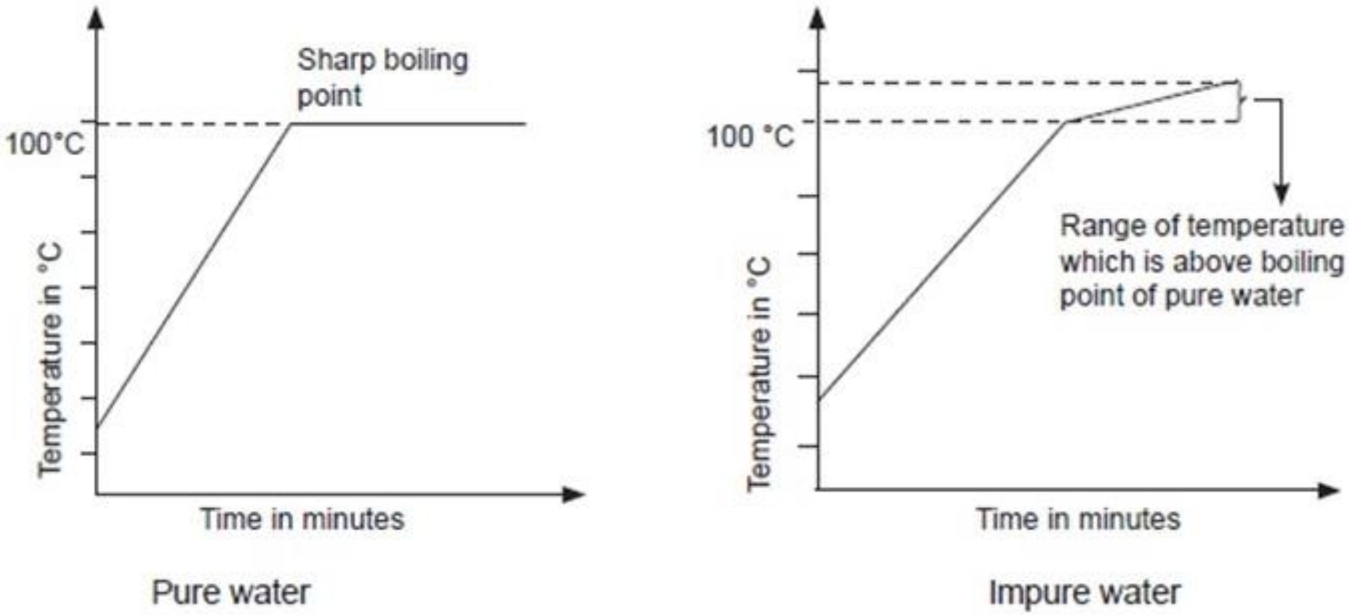
Pure vs. Impure



To understand this, let us go back to ideal solutions!

(Image: https://teacher.co.ke/simple-classification-of-substances-form-1-chemistry-notes/)

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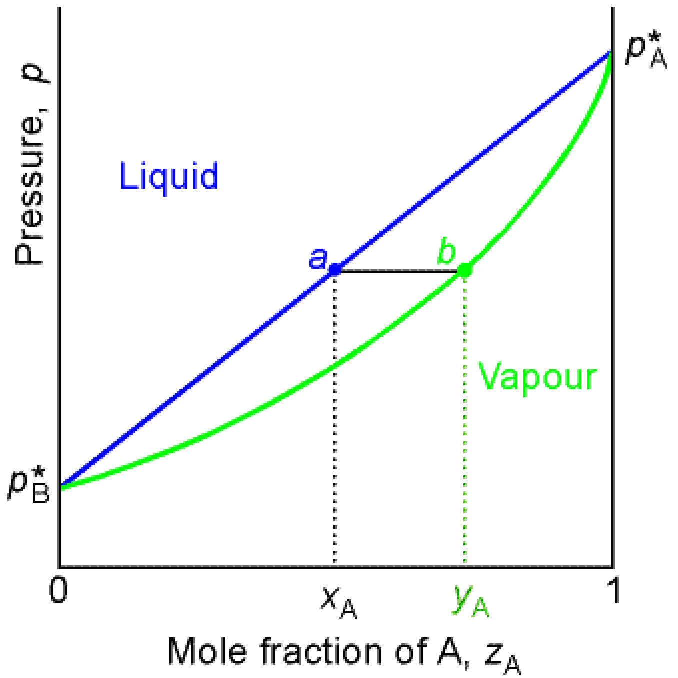
Pressure and Composition



$$p = p_B^* + (p_A^* - p_B^*)x_A$$

$$p = \frac{p_A^* p_B^*}{p_A^* + (p_B^* - p_A^*)y_A}$$

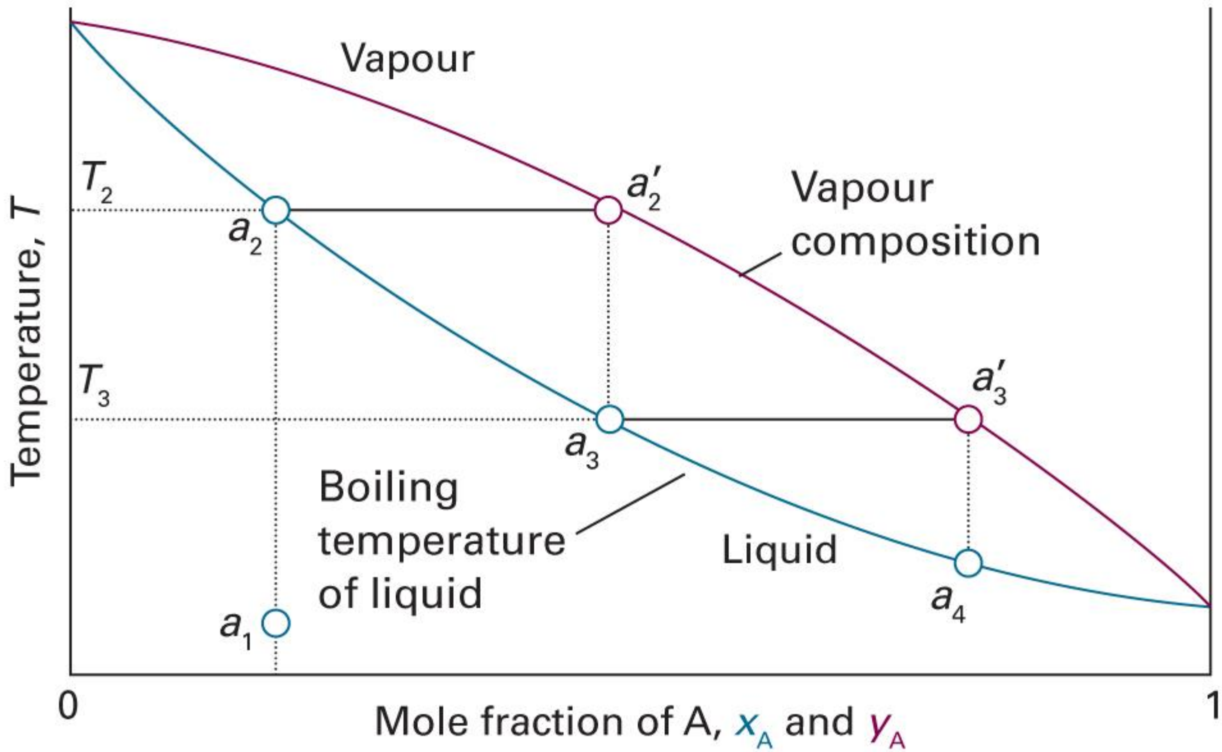
However, p_A^* and p_B^* depend on temperature (Antoine equation):

$$\log p^* = A - \frac{B}{C + T}$$

(Image: http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm)

Temperature and Composition

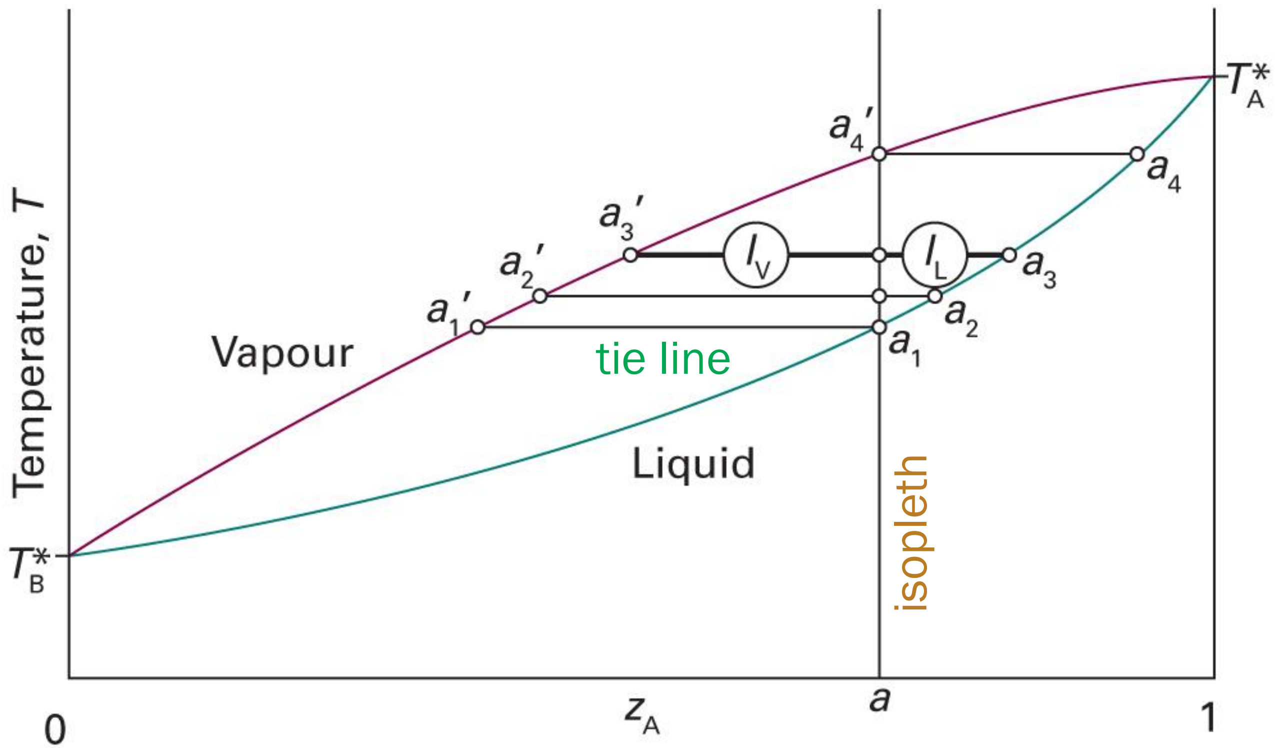
$p_A^* > p_B^*$ (A is more volatile)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.4)

Topic 5C

Interpretation



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.7)

Lever Rule

- Total amount of A molecules: $n_{\mathrm{A}} = n_{\mathrm{A},l} + n_{\mathrm{A},g}$
- Total amount of molecules in liquid: $n_l = n_{\mathrm{A},l} + n_{\mathrm{B},l}$
- Total amount of molecules in gas: $n_g = n_{\mathrm{A},g} + n_{\mathrm{B},g}$

$$n_{\mathrm{A},l} = x_{\mathrm{A}} n_l,\ n_{\mathrm{A},g} = y_{\mathrm{A}} n_g$$

Hence,

$$n_{\mathrm{A}} = x_{\mathrm{A}} n_l + y_{\mathrm{A}} n_g$$

Lever Rule

$$n_{\mathrm{A}} = x_{\mathrm{A}} n_l + y_{\mathrm{A}} n_g$$

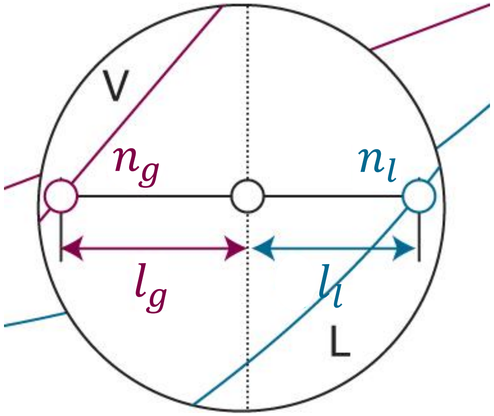
On the other hand,

$$n_{\mathrm{A}} = n z_{\mathrm{A}} = z_{\mathrm{A}} n_l + z_{\mathrm{A}} n_g$$

Equating the two equations,

$$n_l (x_{\mathrm{A}} - z_{\mathrm{A}}) = n_g (z_{\mathrm{A}} - y_{\mathrm{A}})$$

$$n_l l_l = n_g l_g$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.8)

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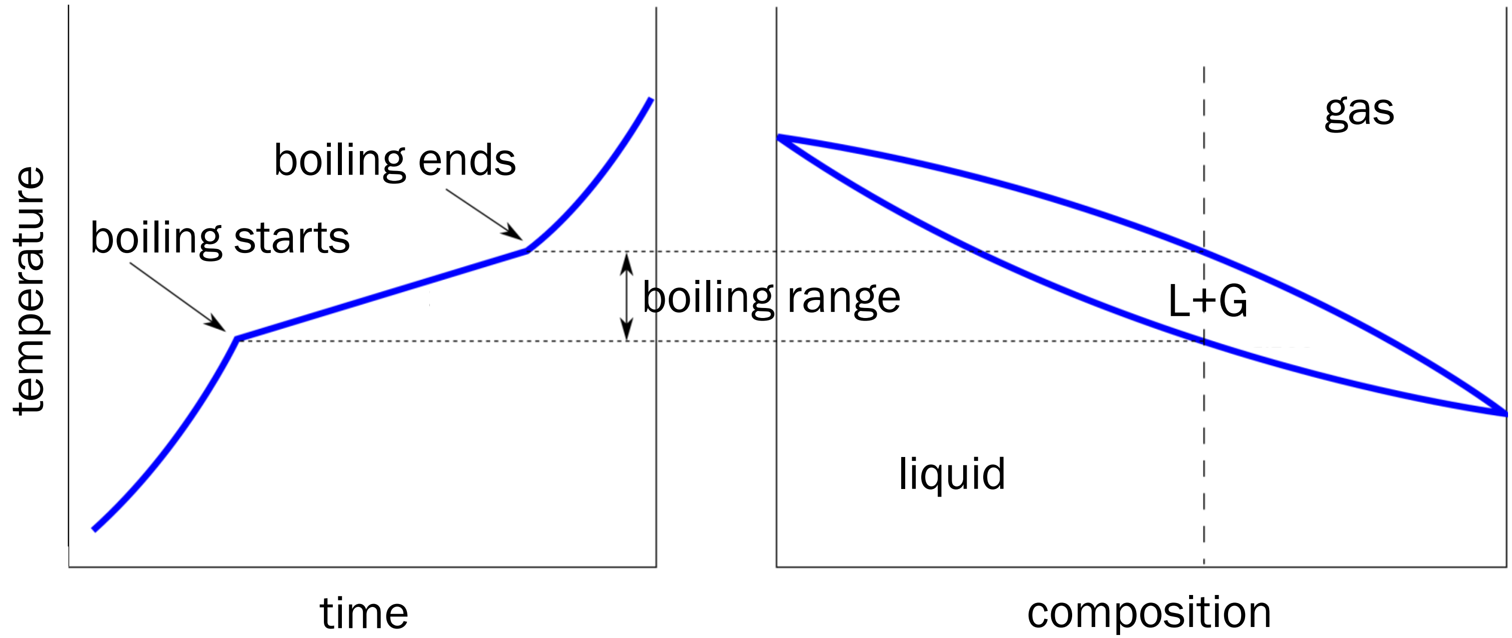
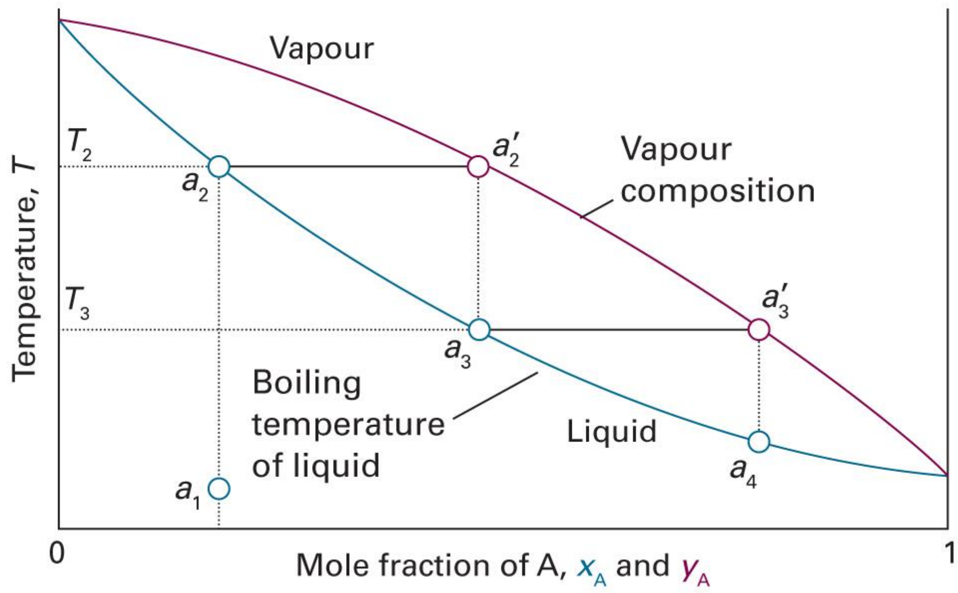
Boiling Curve



(Image: https://commons.wikimedia.org/wiki/File:Cooling_curve_alloy.svg)

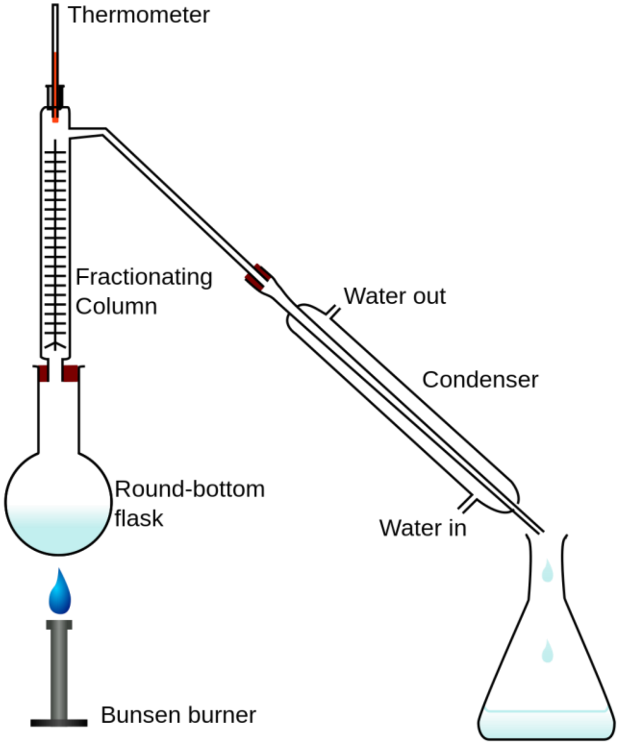
Distillation

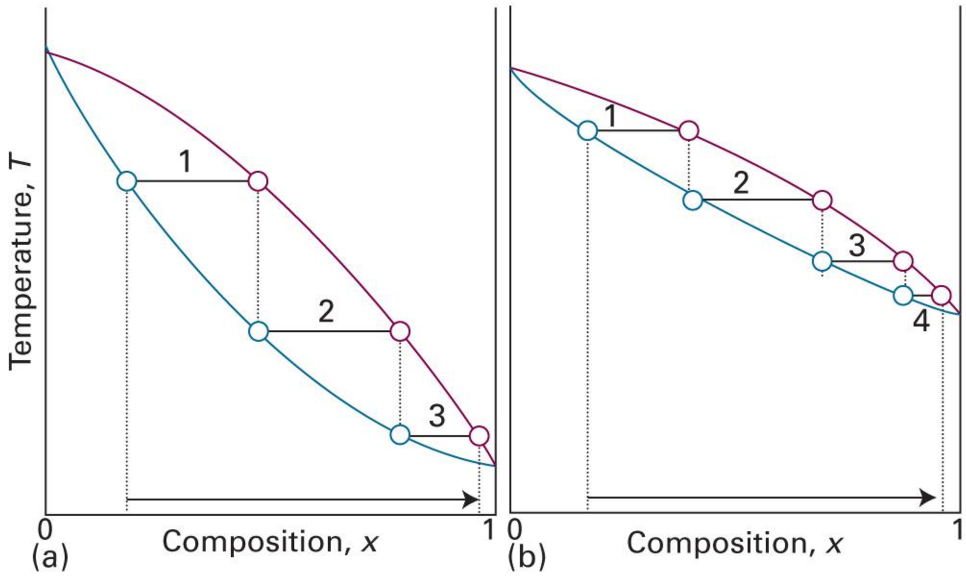
- Simple distillation: vapor withdrawn and condensed



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.4)

Fractional Distillation





(Image: https://commons.wikimedia.org/wiki/File:Fractional_distillation_lab_apparatus.svg
Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.9)

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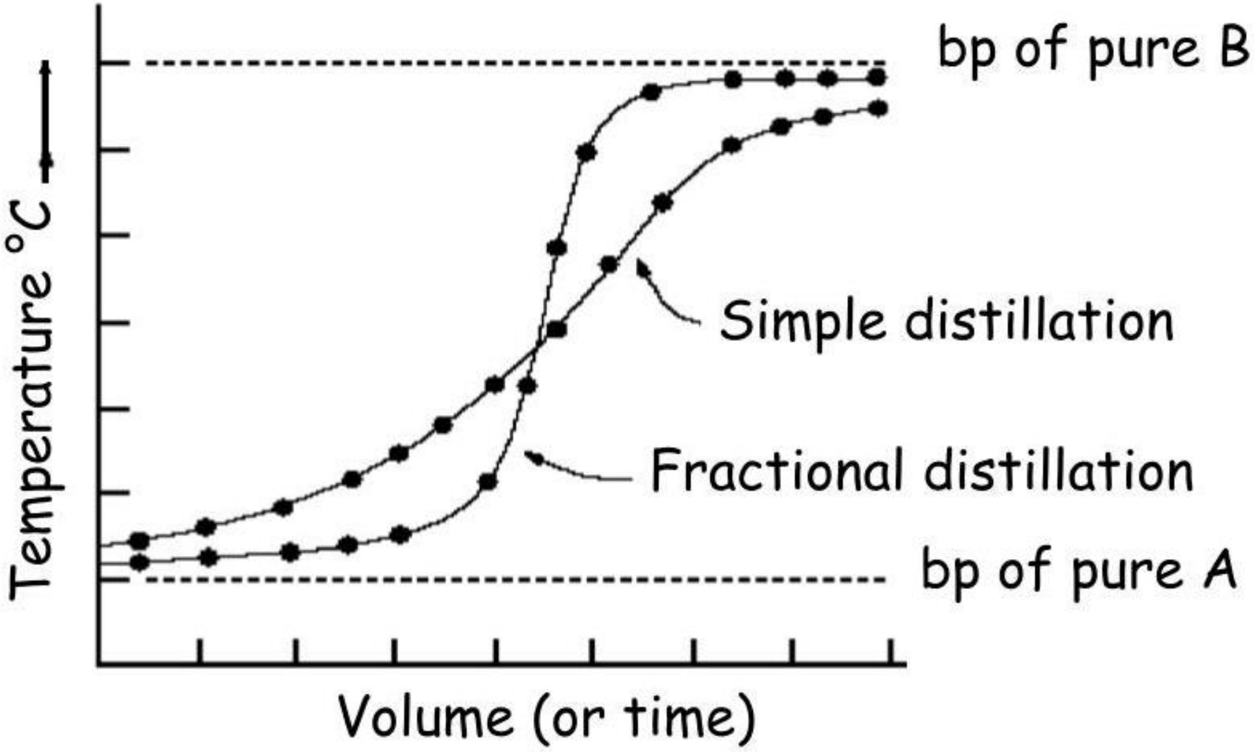
Simple vs. Fractional



(Image: http://www.chem.ucla.edu/~bacher/Specialtopics/Distillation.html)

Real Solutions

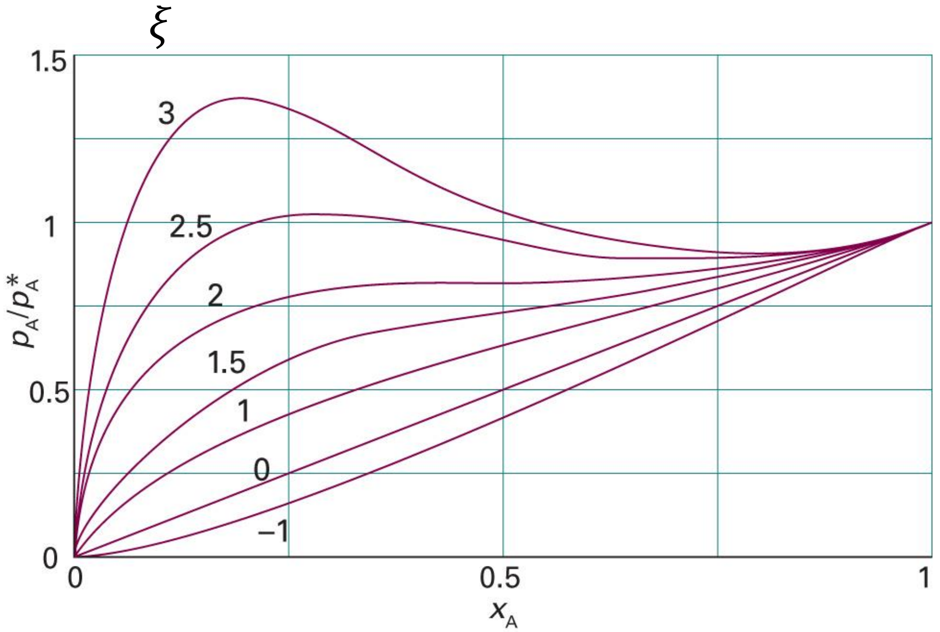
$$\xi = \frac{z}{k_B T}\left(w_{AB} - \frac{w_{AA} + w_{BB}}{2}\right) \neq 0 \quad \text{(BW model)}$$

- $\xi > 0$: AB interactions are weaker
- $\xi < 0$: AB interactions are stronger

1. Change in liquid-gas equilibrium: azeotrope
2. Change in liquid-liquid equilibrium: phase separation

Vapor Pressure of BW Model

$$p_\mathrm{A}/p_\mathrm{A}^* = x_\mathrm{A} e^{\xi(1-x_\mathrm{A})^2}$$

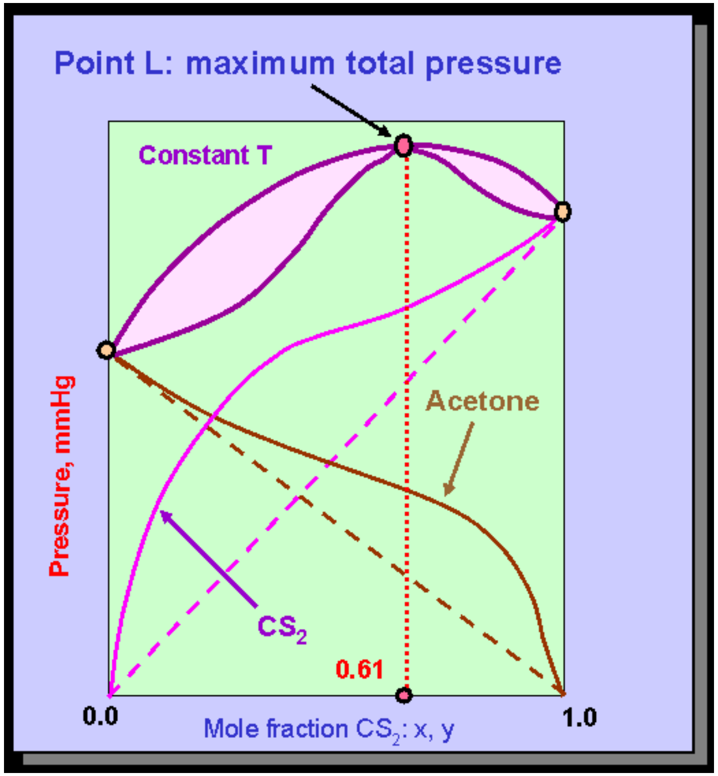


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5F.2)

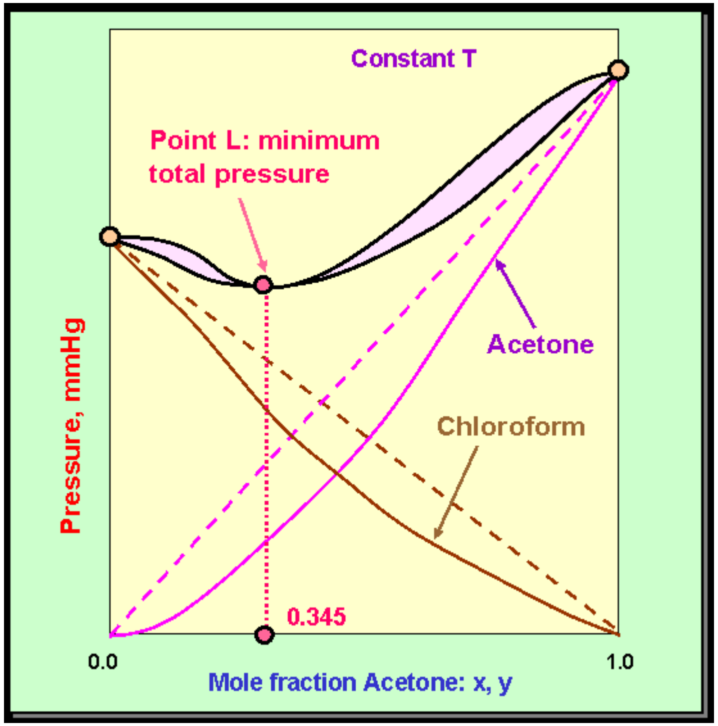
Topic 5C

Non-Monotonic Behavior of p_{tot}

$\xi > 0$



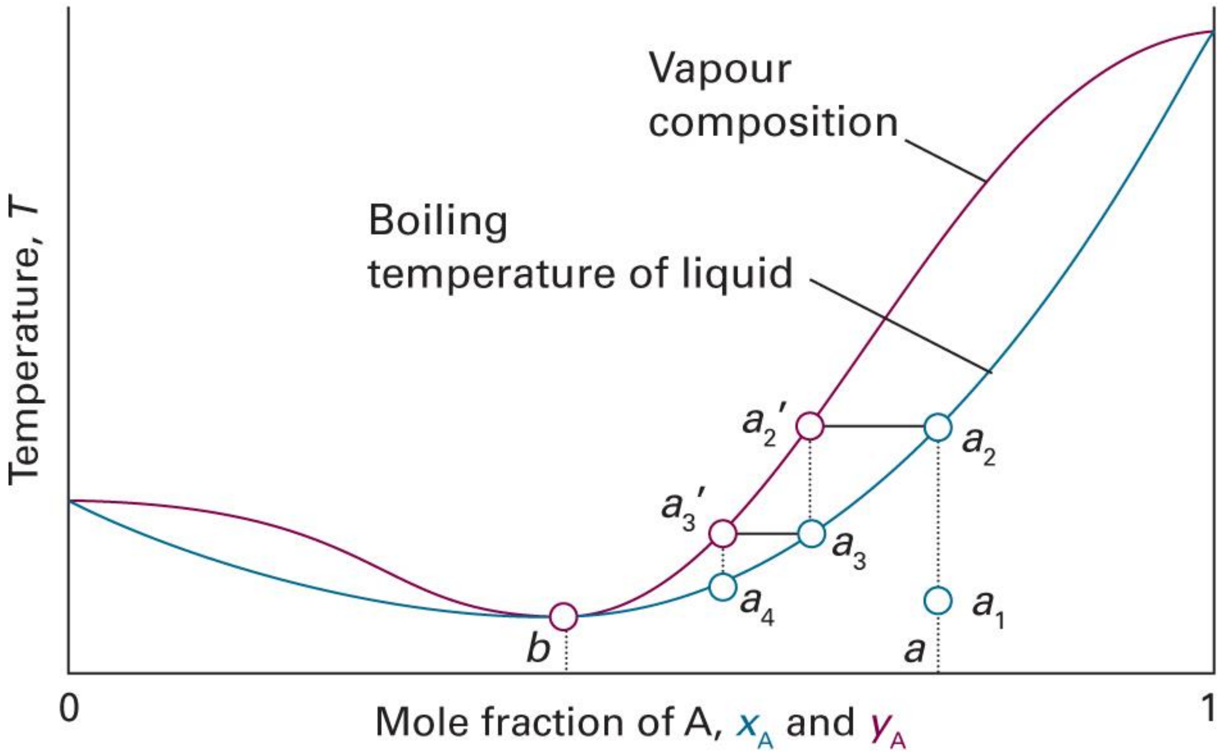
$\xi < 0$



(Image: http://www.separationprocesses.com/Distillation/DT_Chp01f.htm)

Low-Boiling Azeotrope

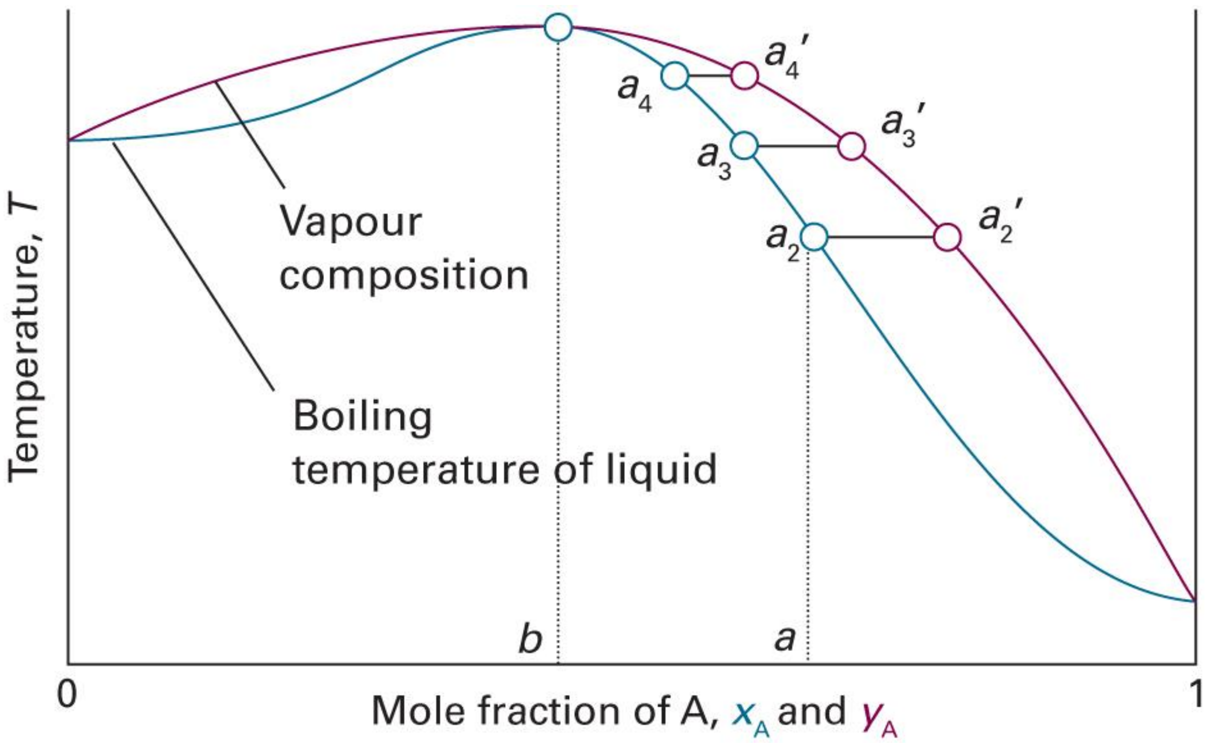
$(\xi > 0)$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.11)

High-Boiling Azeotrope

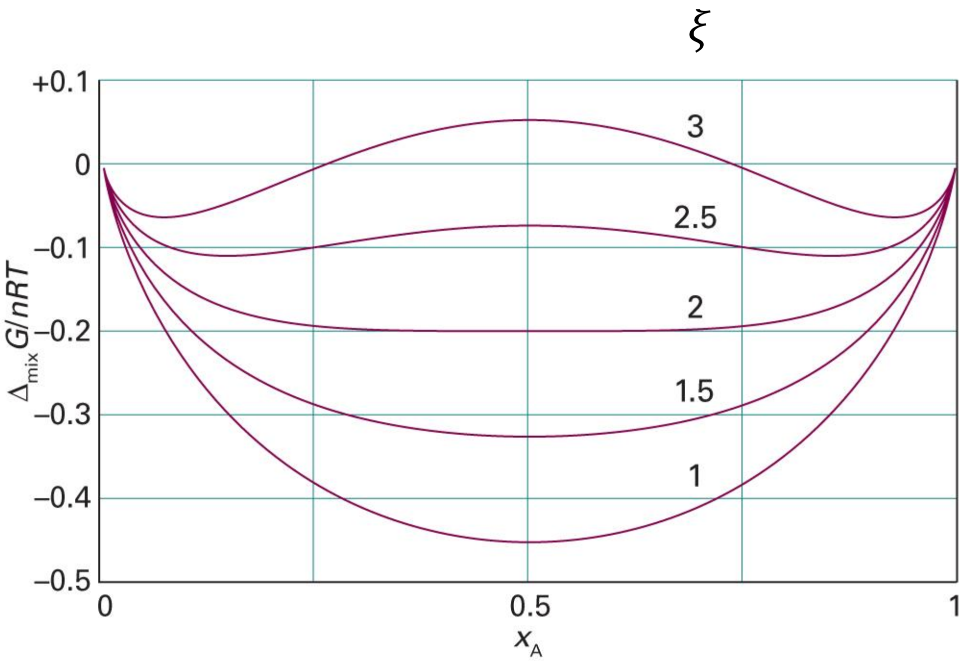
$(\xi < 0)$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.10)

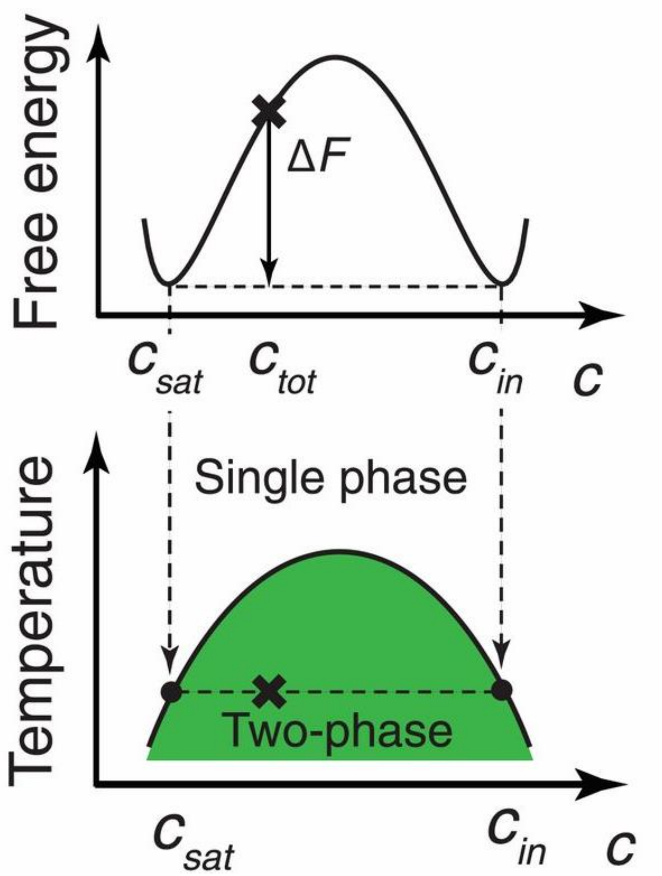
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}} + \xi x_{\text{A}} x_{\text{B}})$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, ***Science* 357**: eaaf4382 (2017), Figure 3)

Locations of Minima

$$\begin{aligned}\Delta_{\text{mix}}G &= nRT(x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}} + \xi x_{\text{A}}x_{\text{B}}) \\ &= nRT\{x_{\text{A}}\ln x_{\text{A}} + (1-x_{\text{A}})\ln(1-x_{\text{A}}) + \xi x_{\text{A}}(1-x_{\text{A}})\}\end{aligned}$$

The local minima can be found by using the derivative:

$$\left(\frac{\partial \Delta_{\text{mix}}G}{\partial x_{\text{A}}}\right)_{T,p}$$

$$= nRT\{\ln x_{\text{A}} + 1 - \ln(1-x_{\text{A}}) - 1 + \xi(1-2x_{\text{A}})\}$$

$$= nRT\left\{\ln\frac{x_{\text{A}}}{1-x_{\text{A}}} + \xi(1-2x_{\text{A}})\right\} = 0$$

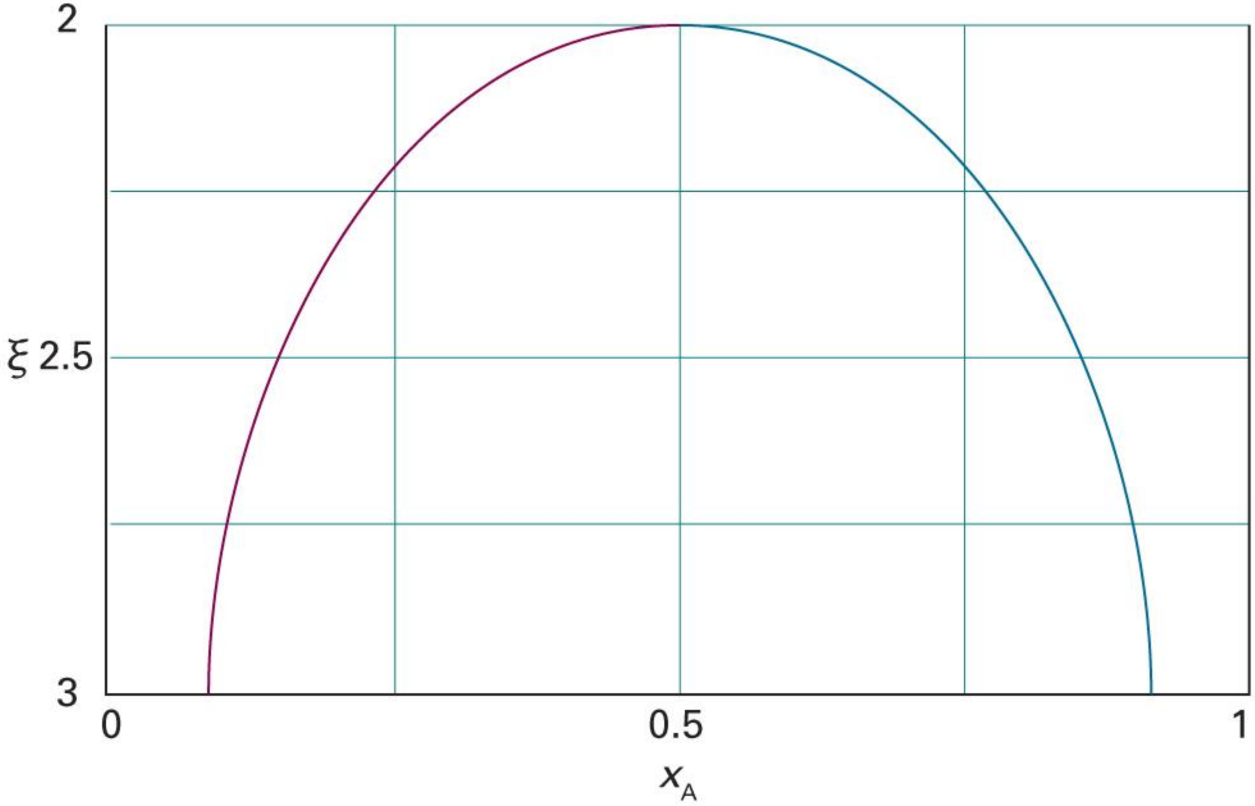
Locations of Minima

$$nRT\left\{\ln\frac{x_\mathrm{A}}{1-x_\mathrm{A}}+\xi(1-2x_\mathrm{A})\right\}=0$$

The minima are located at x_A such that

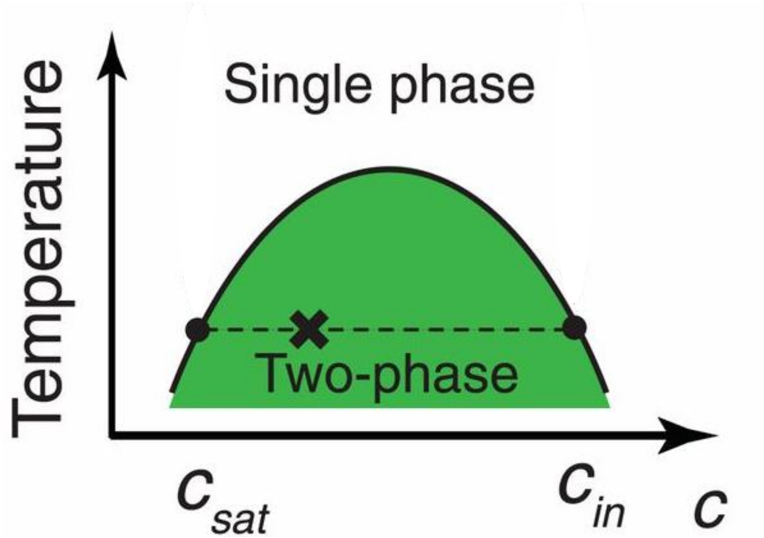
$$\ln\frac{x_\mathrm{A}}{1-x_\mathrm{A}}=-\xi(1-2x_\mathrm{A})$$

Locations of Minima



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.17)

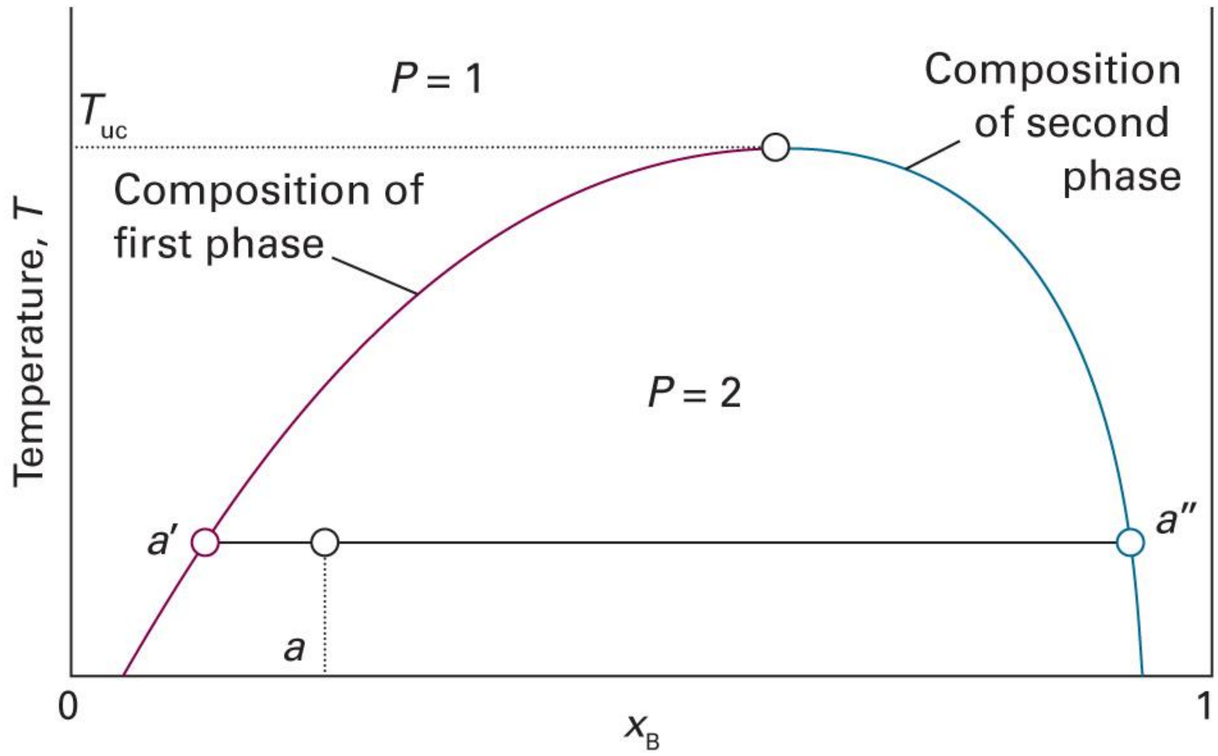
Phase Diagram



Lever rule applies in the two-phase region!

(Image: Shin and Brangwynne, ***Science*** **357**: eaaf4382 (2017), Figure 3)

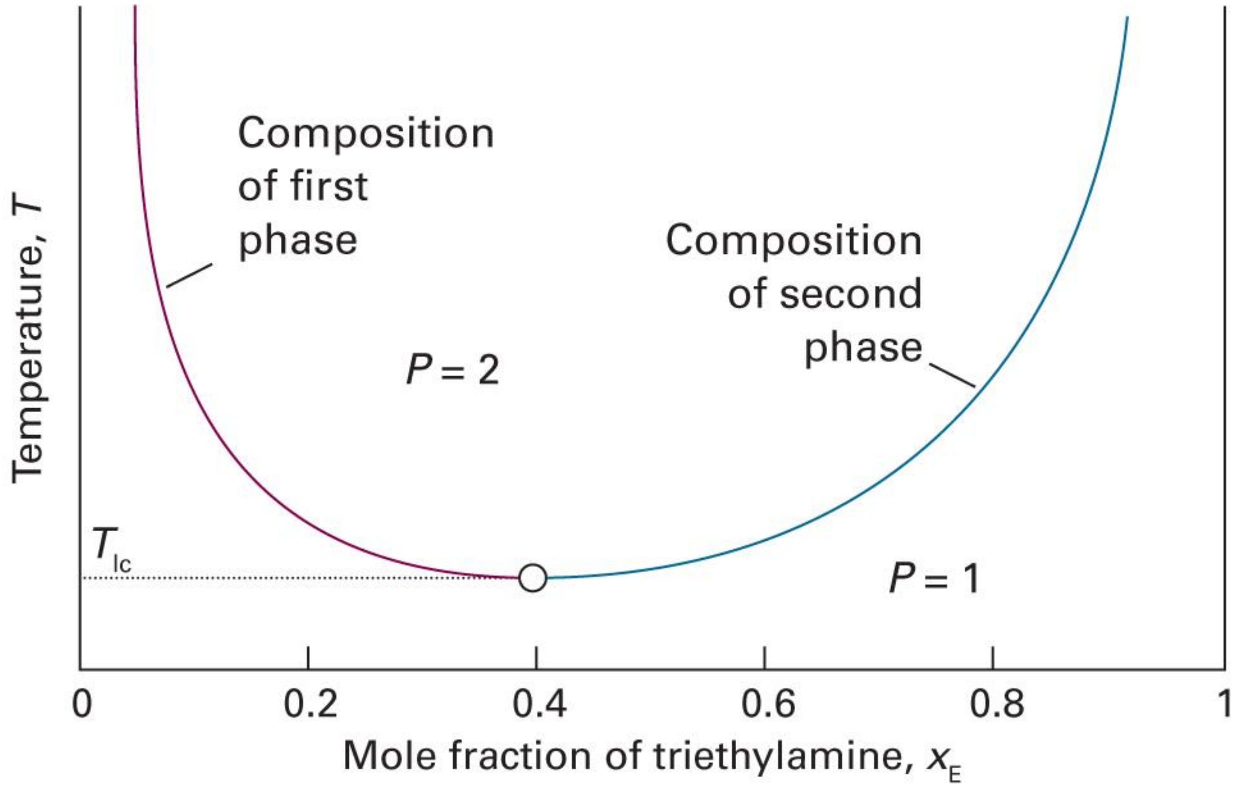
Upper Critical Solution Temperature



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.13)

Lower Critical Solution Temperature

water + triethylamine

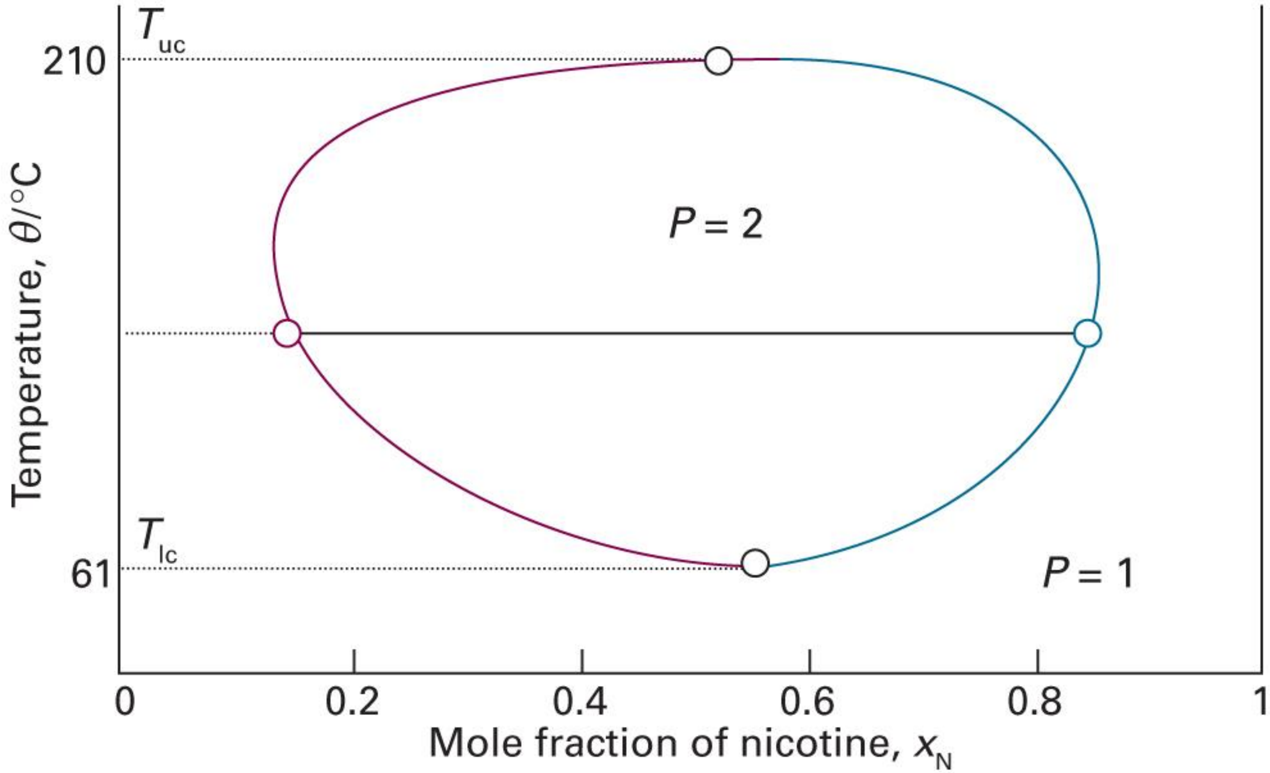


Two components form weak complexes at low T

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.18)

Both UCST and LCST

water + nicotine



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.19)

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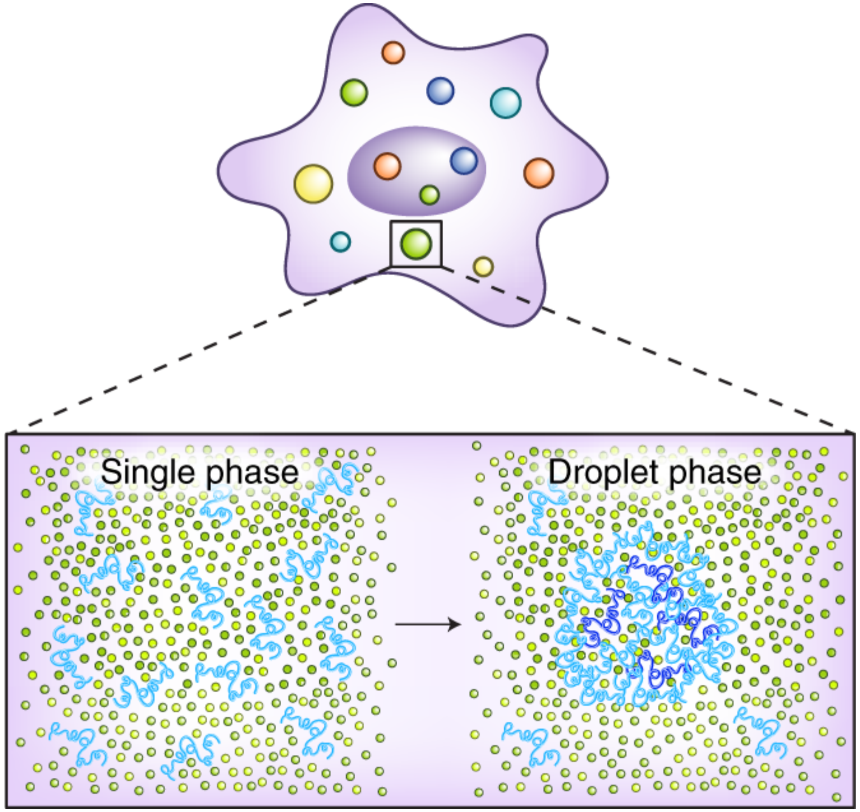
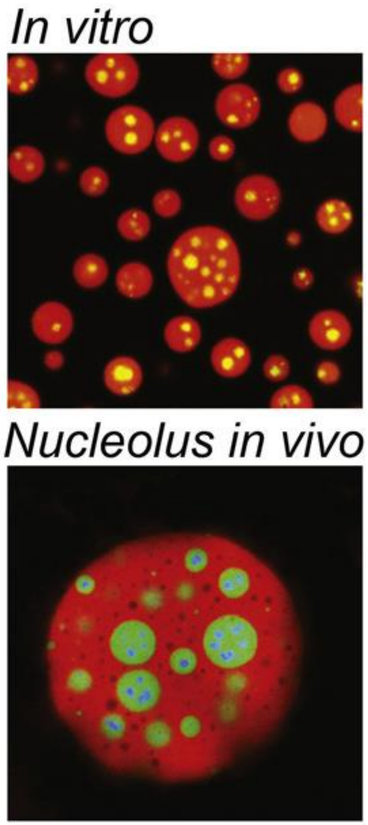
Phase Separation and Life

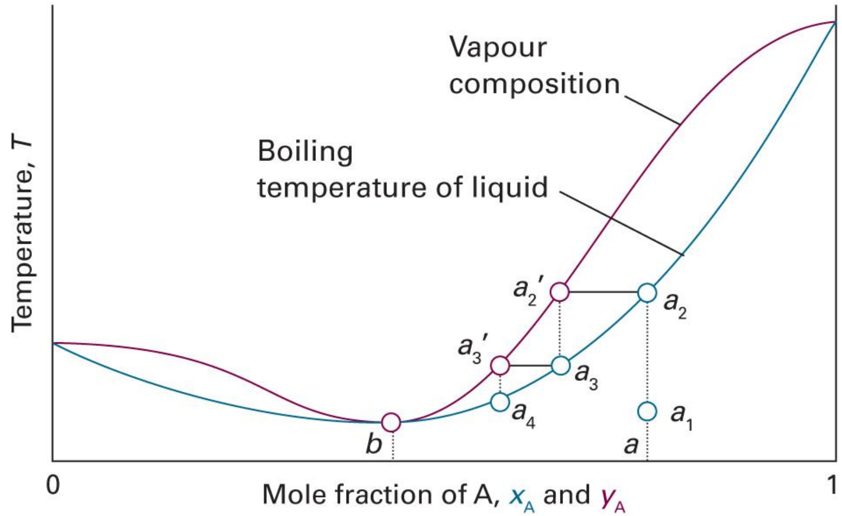




(Image: Tang, ***Nature Methods*** (2019); Feric *et al*., ***Cell*** (2016))

Azeotrope + Phase Separation

(when $\xi \gg 0$)



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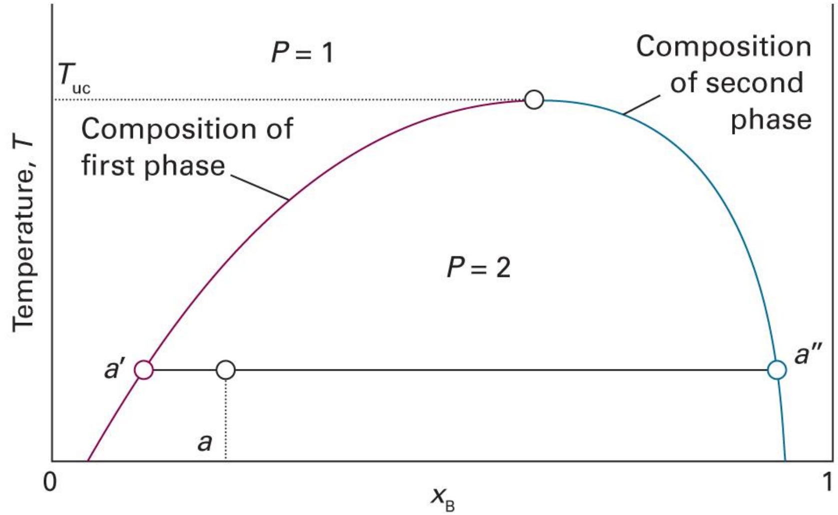
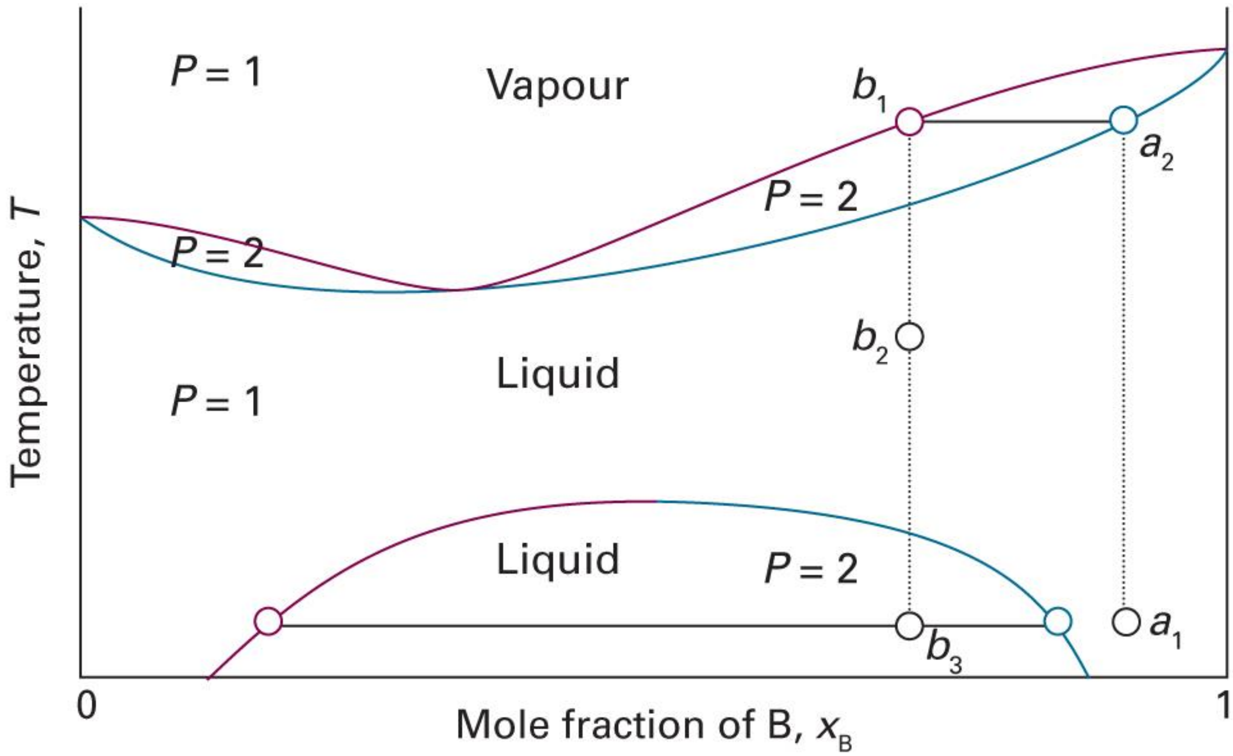


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 5C.11 and 5C.13)

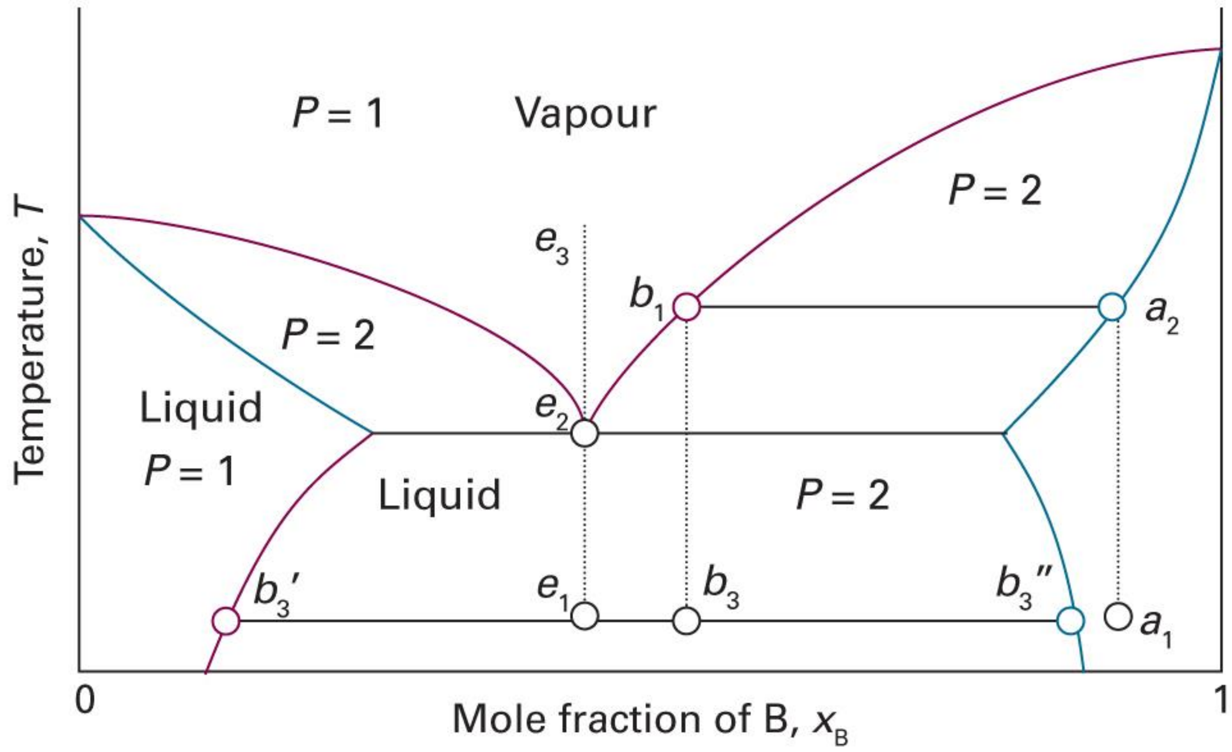
Case 1: Miscible before Boiling



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.20)

Topic 5C

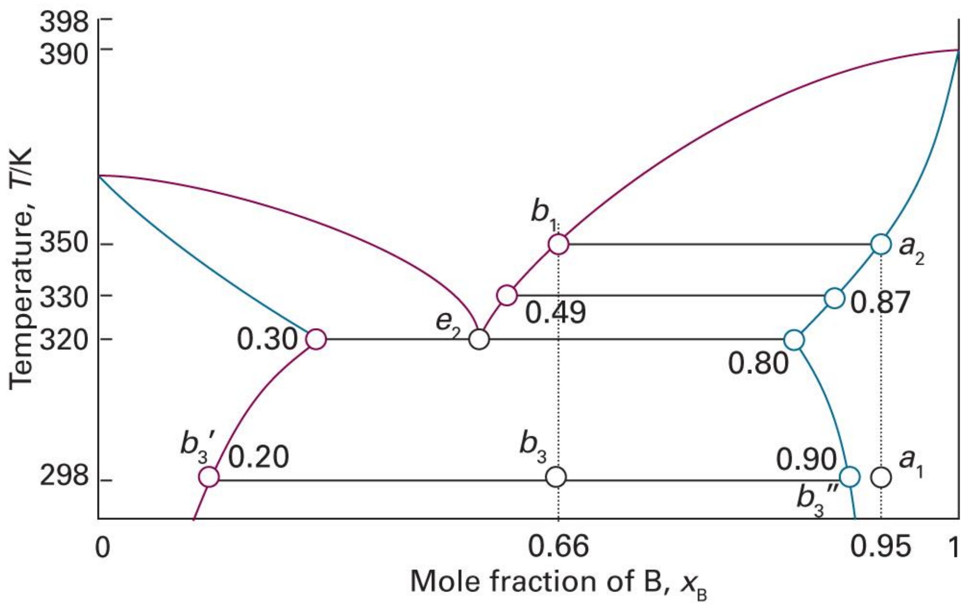
Case 2: Boiling before Miscible



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.21)

Example 5C.3

State the changes that occur when a mixture of composition x_B = 0.95 (a_1) in the figure below is boiled and the vapour condensed.



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.22)

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Liquid-Solid Equilibrium

Various scenarios are possible:

1. Miscible liquid + miscible solid
2. Miscible liquid + immiscible solid
3. Miscible liquid + completely immiscible solid

We can also consider miscibility of liquid

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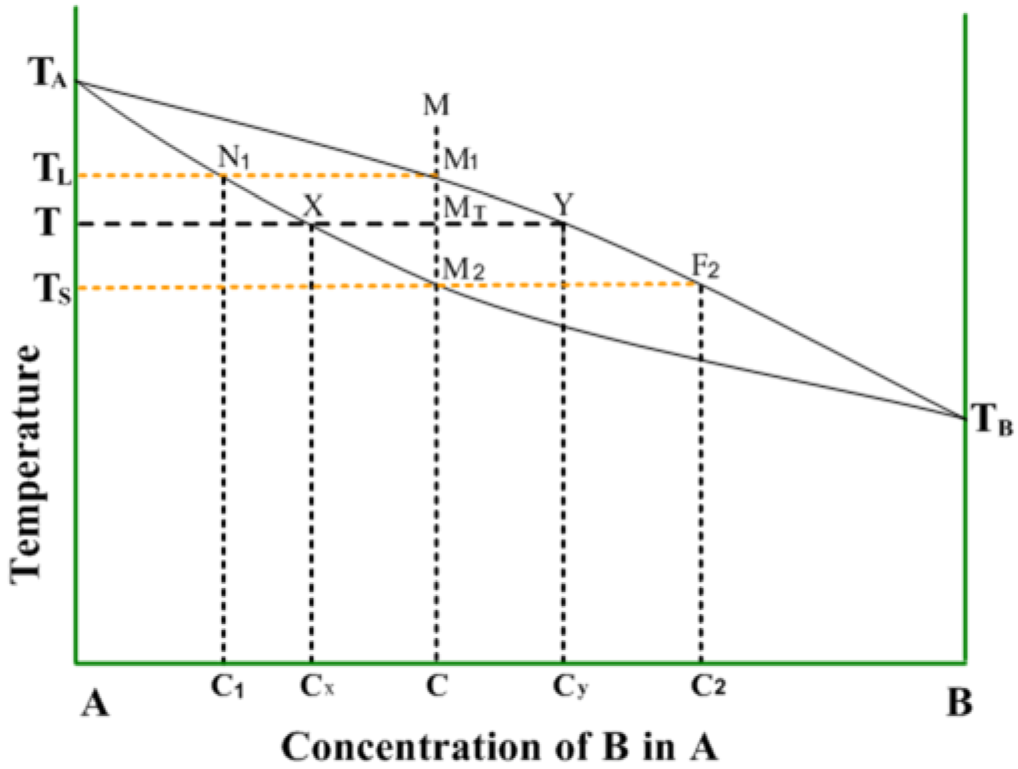
Miscible Solid



(Image: https://www.substech.com/dokuwiki/doku.php?id=phase_transformations_and_phase_diagrams)

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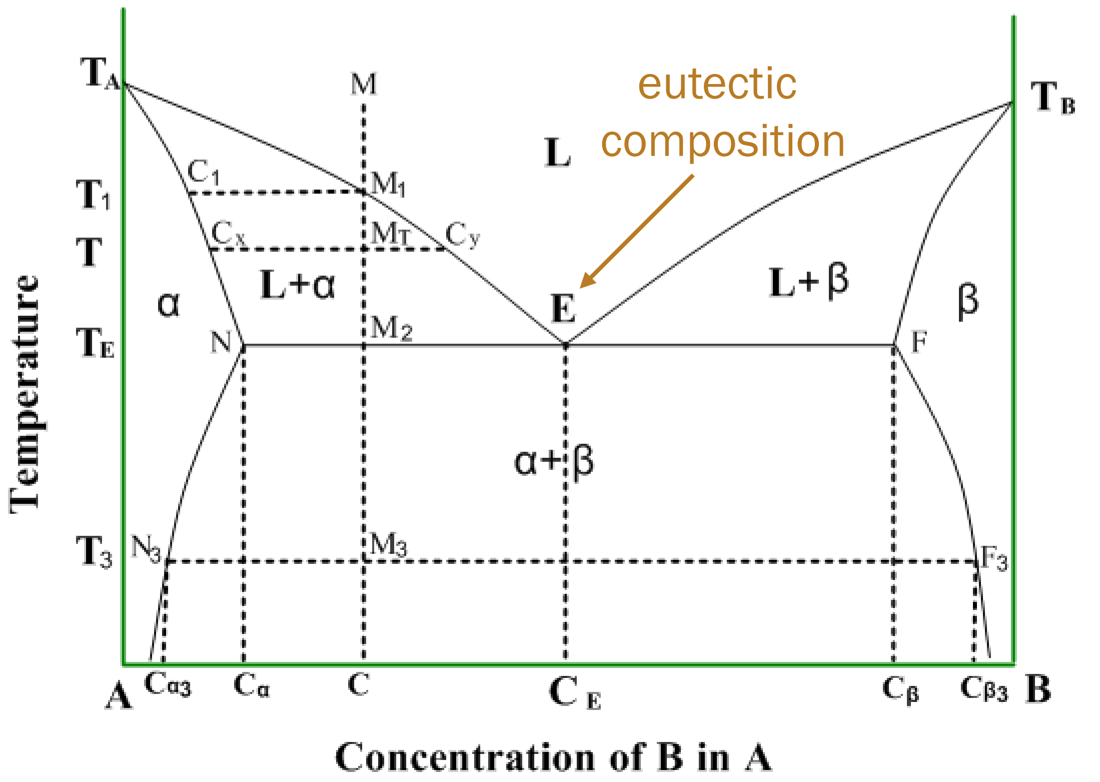
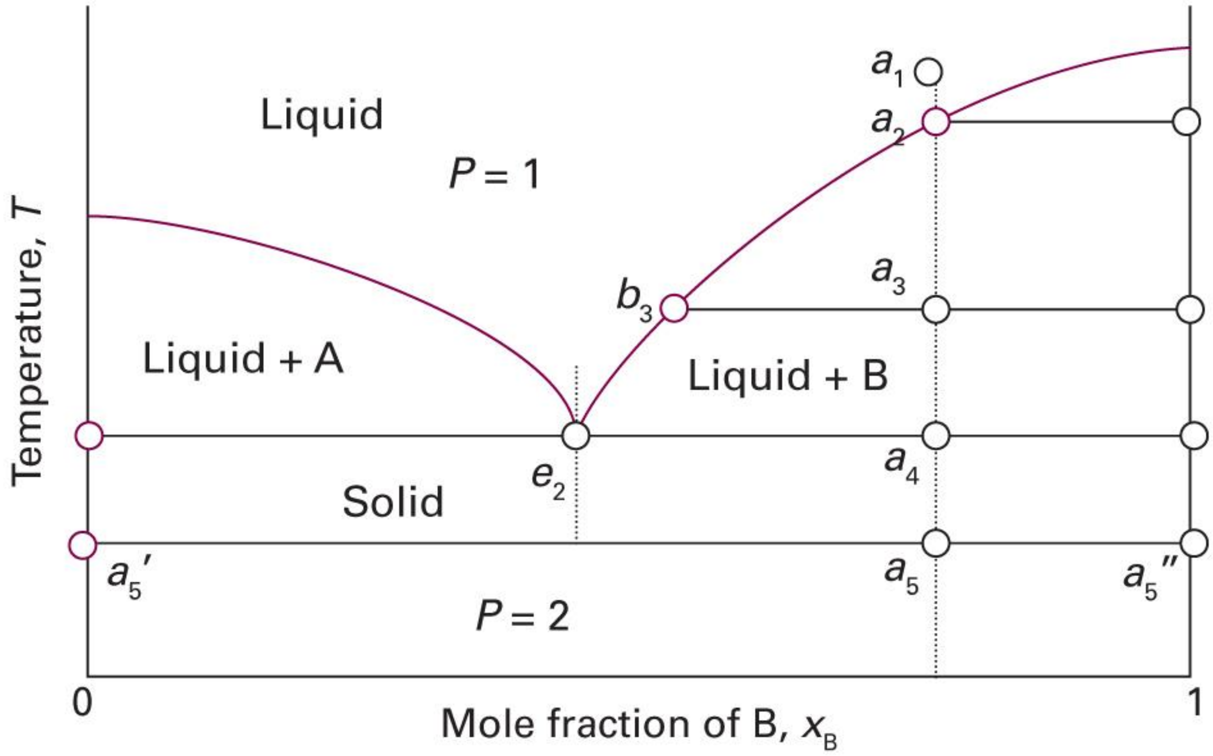
Immiscible Solid



(Image: https://www.substech.com/dokuwiki/doku.php?id=phase_transformations_and_phase_diagrams)

Completely Immiscible Solid



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5D.1)

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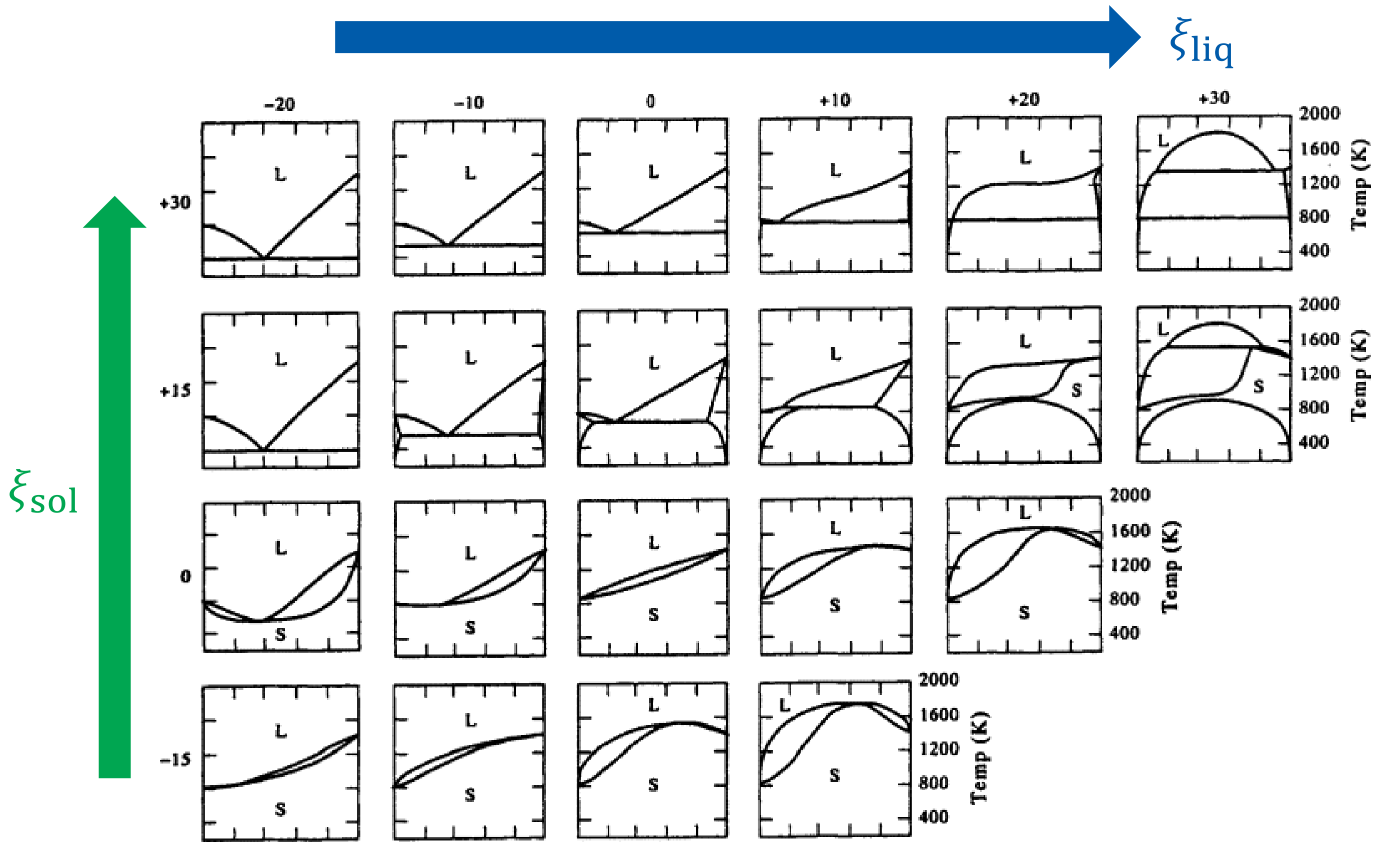
Diverse Phase Diagrams



(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

Reacting Systems

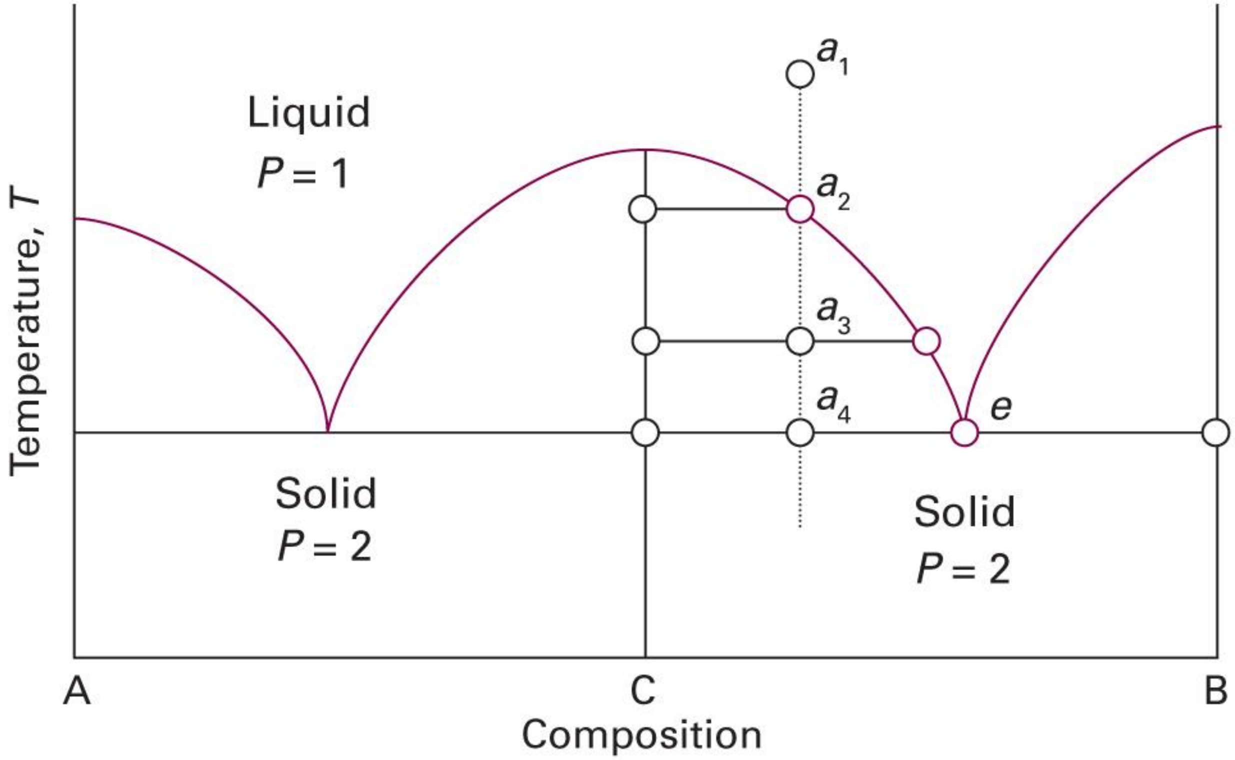
For reaction A + B ⟶ C (irreversible),

- A 100%, B 0% = A only
- A 75%, B 25% = A 2/3, C 1/3
- A 50%, B 50% = C only
- A 25%, B 75% = B 2/3, C 1/3
- A 0%, B 100% = B only

The x-axis can be (A ⟶ C ⟶ B)

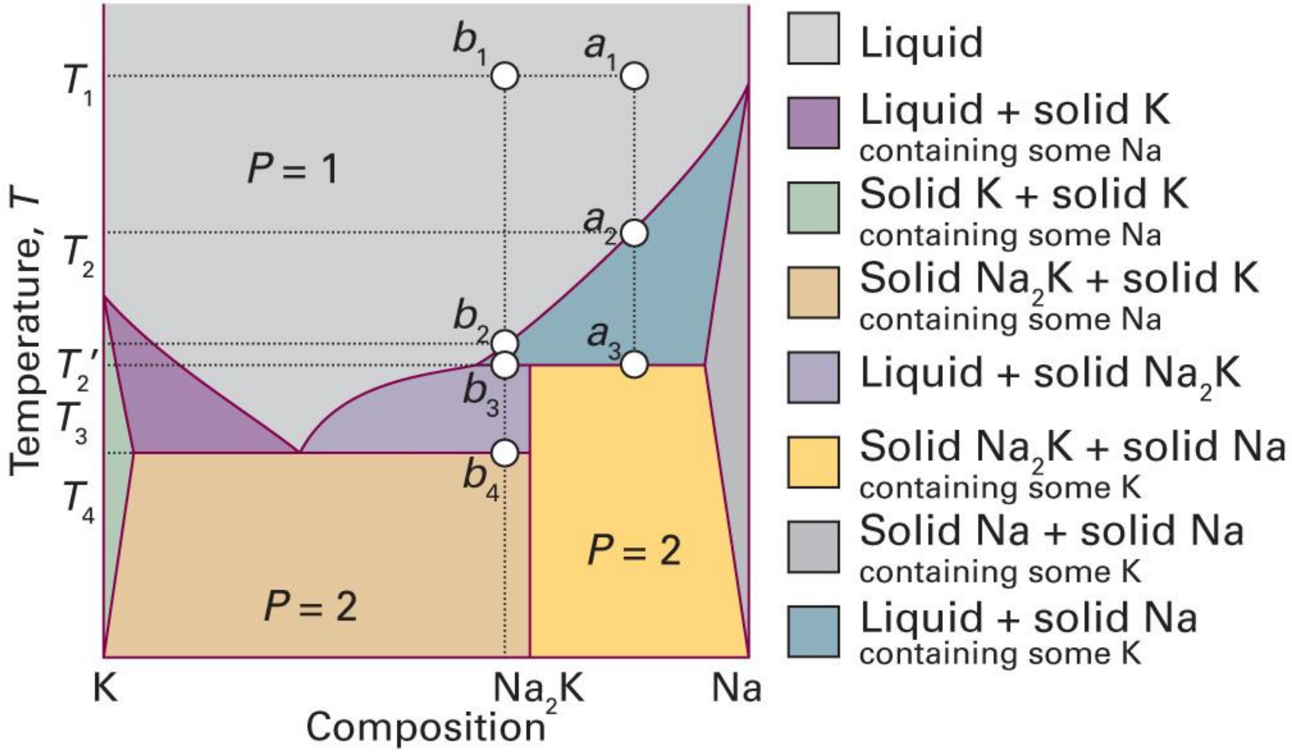
Congruent Melting

(assuming completely immiscible solid)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5D.4)

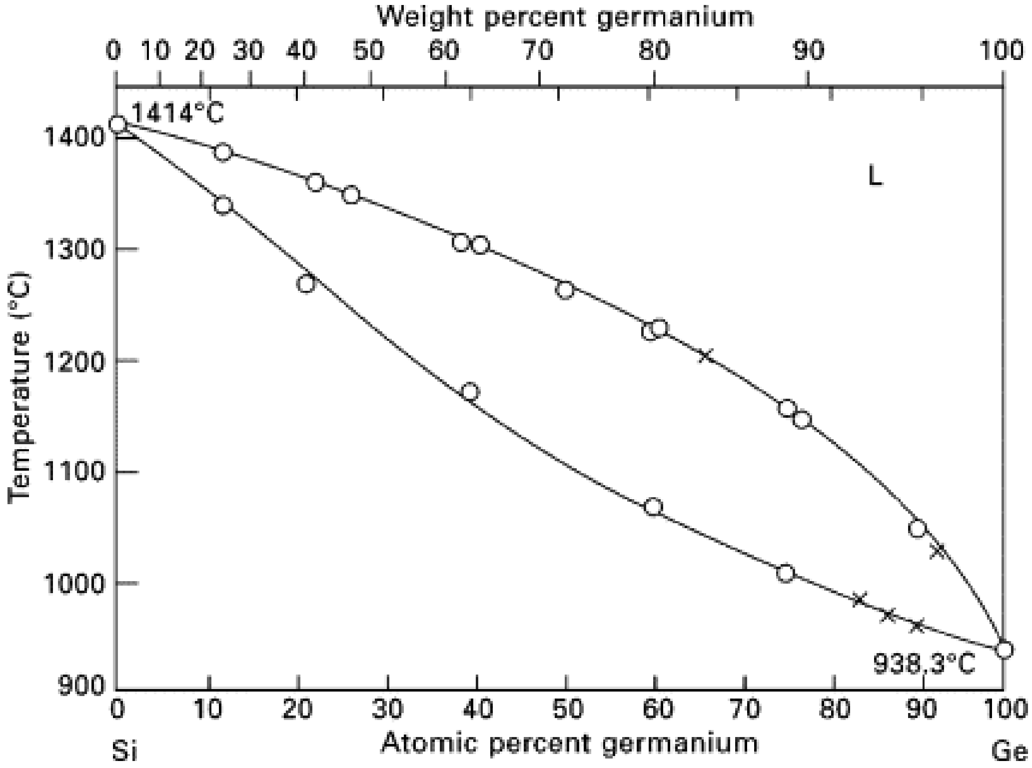
Incongruent Melting



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5D.5)

Not in Atkins

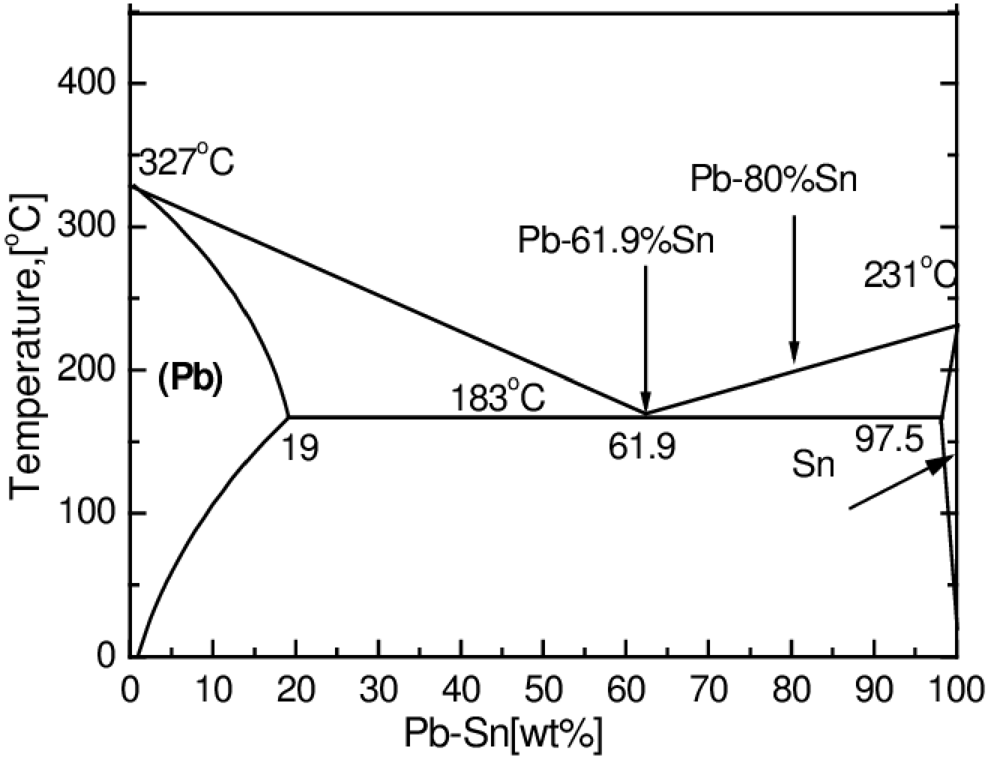
Si-Ge System



(Image: Kasper and Herzog, ***Silicon–Germanium (SiGe) Nanostructures*** (2011), Fig. 1.2)

Not in Atkins

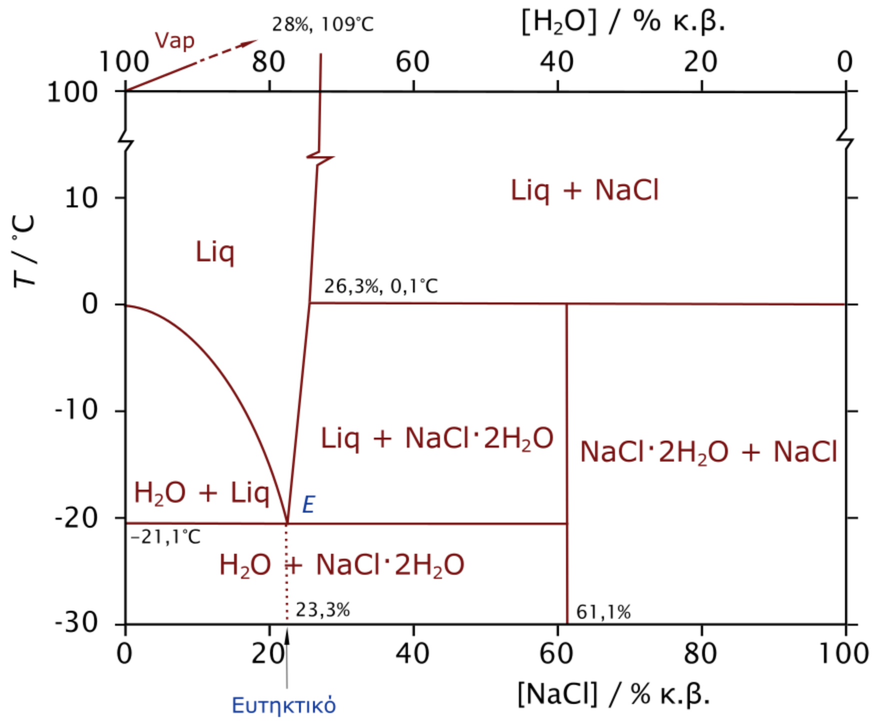
Pb-Sn System

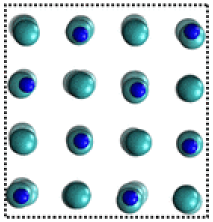


(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

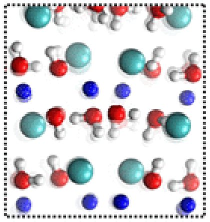
Not in Atkins

H_2O-NaCl System

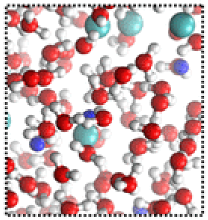




NaCl



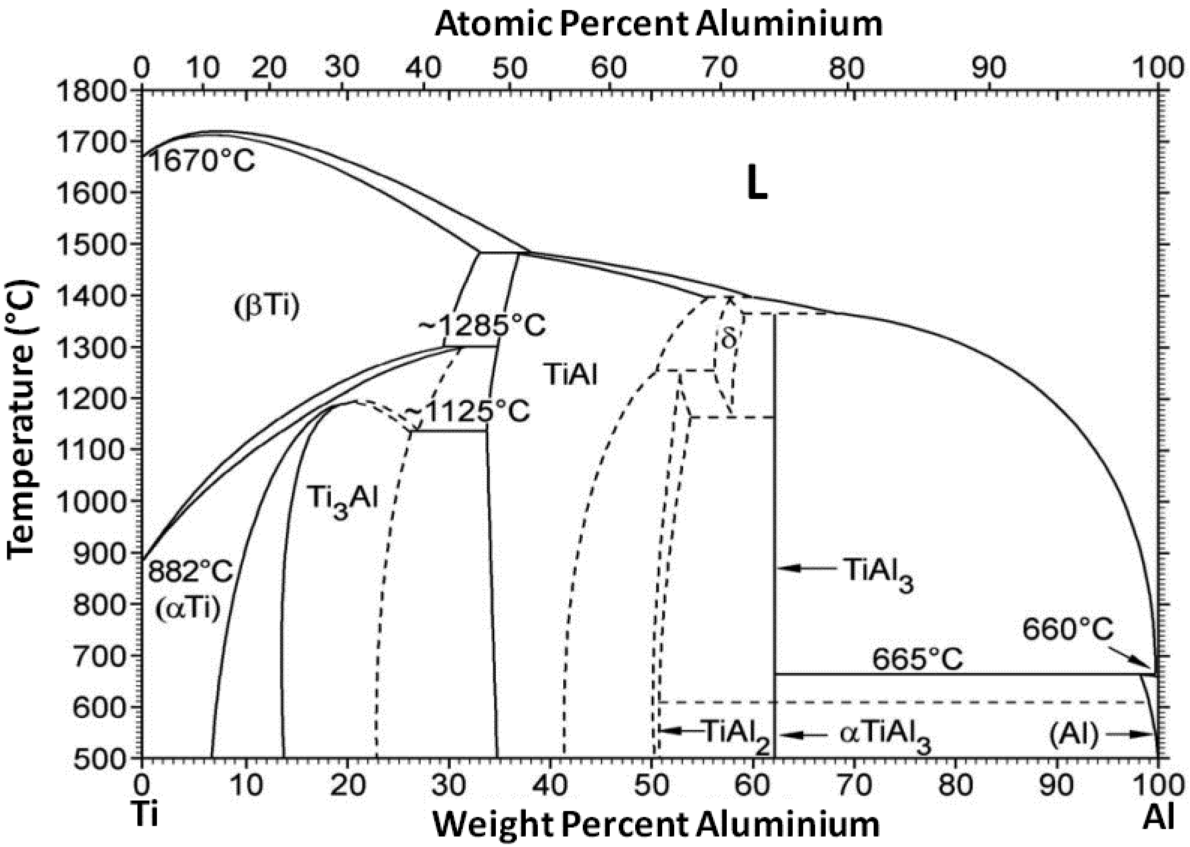
$NaCl \cdot 2H_2O$



Liq

Not in Atkins

Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)